# When Does Admission Control Pay? The Memory Tier Sets the Cost of Concurrency in Agentic LLM Serving

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***Abstract*—Agentic LLM sessions are long, multi-round jobs whose KV footprint grows every round, and on a memory-constrained accelerator the pool is the binding resource. Admission control therefore looks like the obvious lever: a controller that reads engine telemetry and admits on measured capacity should beat a static cap. We give a measurement that says how much room there is, and show what happens when you build against it. Throughput during busy time is $\overline{R}/\tau$—running requests over per-step cost—so raising concurrency by a factor $f$ buys $f^{1-\epsilon}$, where $\epsilon = \mathrm{d}\ln\tau/\mathrm{d}\ln\overline{R}$. Near zero, busy-time throughput keeps converting; near one it does not. It is a property of the workload *and* the engine configuration, not of the controller, and what sets it is the memory tier. The trace bounds reuse at 75.5% of its prompt tokens, and no configuration we ran reaches that bound—the best realizes 92.5% of it. What the tier decides is how much of the bound survives added concurrency: device-only, realization falls from 92.5% to 36.4% as the running count rises from 1.00 to 1.84 and the shortfall appears as recomputation (8.6 M to 20.7 M tokens); with a host tier the static points hold at 90.2–92.4% between 1.25 and 1.73, and across the nineteen host-tier runs that completed this trace realization never falls below 72.8%, while the *device* cache misses more. Two configurations at the same running count and the same pool occupancy realize 79.0% and 91.3% of the bound, so it is not how full the pool is but what happens when it overflows—though that pair is two runs on two days, so we also have a capacity control in the queue that changes the pool size alone on one card; we report it when it lands. Our controllers run on the host-tier configuration, where $\epsilon$ stays below one across their whole operating range, so the ceiling we measure on device-only caps is not what bounded them: it bounds what added concurrency is worth, not what a whole run achieves—idle time and the SLO axis are separate. We measure $\epsilon$ on a memory-bound agentic replay (200 sessions, one 24 GB GPU, a pool that holds four median sessions) and get $0.38$ to $0.61$ depending on which step cost a system can observe: every 10% of added concurrency buys 4–6% of throughput. A second, real agent trace (117 Claude sessions, same engine and pool) reproduces both measurements in direction, neither in level: realization falls there too, and the elasticity there also turns positive—rising to 1.0 where the main trace's reaches 1.15—at a different running count and by an amount our own perturbation study cannot pin down.**

**The verdict on the controllers is negative, and we report it at that strength. (The criterion of §IV is a diagnostic; that section is explicit that it neither returns this verdict nor explains the failure.) Nine controller variants spanning four routes, plus a reimplementation of an independent congestion-based controller, none is above the static Pareto front at the latency threshold interactive serving cares about—of the six whose measurements are clean; the other three were measured while other jobs shared the host and are being re-measured. The best of the six misses the front by under a point, on the below side of our tie band, and the closest arm we repeated misses by 0.9 with its repeat at 5.6; a plain cap on resident sessions remains the fastest policy we measured. That is not an argument for admitting everything: with no control at all the median request waits 1,774 seconds for its first token, the prefix cache's realized hit rate collapses from the trace's planned 75.5% to 1.6% (both are the runner's own end-of-run summary figures, on its own all-steps convention; `nocontrol.csv` carries them alongside four conventions we compute ourselves, which give 73.8% and 1.9% for the same run), and SLO falls to 3%. The question is which selective rule, not whether to be selective. Two measurable effects, rather than one we can tune away, account for part of why our variants fail—the arm-by-arm attribution of §V-C attributes four of the twelve controller arms we tested, leaves five with a step cost that no matched comparison can confirm, and leaves three as residuals we can measure and cannot yet explain. The single step-cost attribution it does make is on a run flagged for host co-tenancy but not measurably damaged by it, and we say which is which rather than counting it. A fourth flagged variant has been re-measured: it moved from 7.8 points below the front to 1.3, still below. They hold nearly half the pool idle while sessions queue—their own headroom estimate on the base configuration is 1.79× the truth (1.32–1.90× across the sixteen controller runs that finished on this trace, and 1.01–4.11× over every finished controller run we logged)—and the per-step elasticity measured on the configuration they actually run stays below one across their whole operating range, so the engine had room they did not take: the ceiling we measure on device-only caps does not survive on the host-tier configuration the controllers use. The wall clock splits into busy and idle time; admission policy moves only the running count, and $\tau$ rises with it. The idle component (7.3–41.9% of the sampled span) comes from gaps the tool waits set—an agent session issues no request while a tool runs, and that wait averages 16.85 s measured over the controller's run, against the 15.82 s the trace demands—on the static cap of one the two match exactly, so we attribute the controller's extra 1.03 s to its own admission delay—not the sub-second figure the trace's median suggests—though how many such gaps a run collects is partly the policy's, since a policy sets how many sessions stay live. On device-only configurations we can account for the rise token for token; on host-tier ones part of it is co-tenancy we found late and part we still cannot attribute, and we say which is which. We also report the six of our own claims we retracted along the way, and the boundary of the verdict: at a loose latency threshold one variant appears above the front in a single measurement, and that variant's own repeat lands 2.3 points *below* it, so the crossing does not reproduce.**

***Index Terms*—LLM serving, admission control, KV cache, agentic workloads, performance characterization, negative results**

## I. Introduction

An agentic LLM session is not a request. It is a long-lived job that issues a round of inference, waits on a tool, and comes

back—tens of times—with a context that grows every round. On a memory-constrained accelerator this makes admission control look like the obvious lever. The KV pool is the binding resource; a single large session can occupy most of it; and the sessions themselves are wildly heterogeneous. If a scheduler can see how much of the pool a candidate will need and how much room is left, surely it should beat a fixed number.

We believed this, built a controller that does exactly that, and spent a considerable amount of GPU time discovering that on our workload and hardware it is false. This paper reports that negative result, and the negative result is most of what we have. Alongside it we report one measurement—the elasticity of per-step cost with respect to the running count—that we hoped would say in advance how much room admission control has. It does bound that room on the configurations we measured, but when we tested whether it explains our controllers' failure, on their own configuration and on a second workload, it did not. We report it as a diagnostic of a workload, not as a predictor of whether a controller will win.

### A. What we measured

We serve a replay of 200 real coding-agent sessions on a single RTX 3090 (Qwen2.5-Coder-7B, 96K context, SGLang, a 101,432-token KV pool) and compare a family of admission policies against a sweep of static concurrency caps. Nine controller variants span four mechanistically distinct ideas, and a reimplementation of an independent congestion-based controller from the literature runs alongside them. Six of the nine have measurements we can compare, and none of those six is above the static Pareto front (Fig. 1). The best of them misses by under a point—inside the $\pm 1.0$ pp band we call a tie, but on the below side of it—and the closest arm we repeated misses by 0.9 with its repeat at 5.6. Four of the nine were measured while other jobs shared the host; we re-measured one, which moved 6.5 points closer to the front without crossing it, and §V says what the verdict rests on until the other three land.

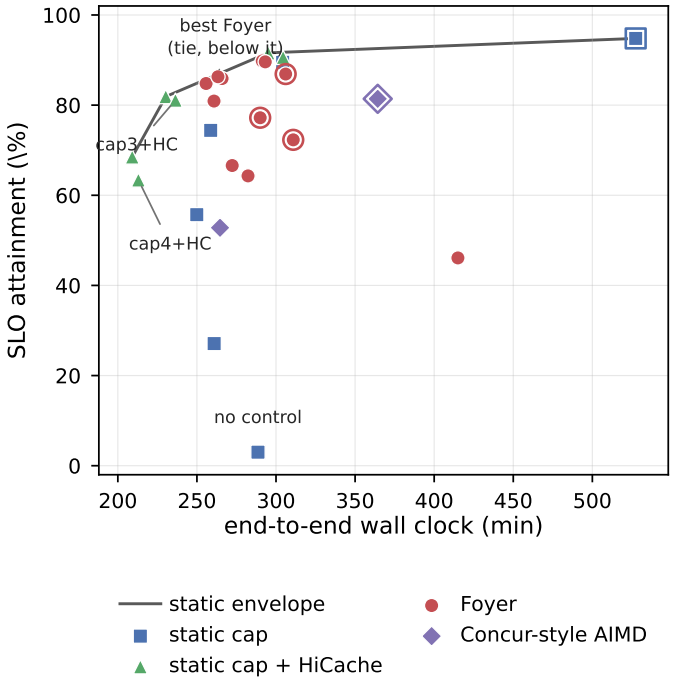



Fig. 1. Every completed run on the (wall clock, SLO) plane. The grey line is the upper-left envelope of the static points—drawn from the data, so it uses each configuration's *better* run where a repeat exists, which is why `cap3`+HiCache's 236-minute run sits below the line rather than on it; a policy below or on it has not beaten the *line*, because time-slicing two caps approximates any point on the segment between them. That is an argument about an interpolation, and we do not have the switching policy that would turn it into a measurement; §II states the limit and names the arm that beats the nearest measured static point on SLO while losing on wall clock. Nine Foyer variants plus a reimplementation of an independent controller, and of the six Foyer variants whose measurements are clean, none is above at the 10 s latency threshold used throughout; §IX reports how far that verdict is conditional on the threshold. The five points drawn with a hollow ring were measured while three or more other engine instances shared the host (§IX-0e); on three of them the per-request decode interval is inflated by 24–41% and they are not comparable with the rest, on the other two the flag is a window that left no measurable trace. The three are being re-measured, and one arm that carried the flag in an earlier version has been re-measured clean and is plotted without a ring.

### B. Why

We decompose the wall clock into *busy* time, during which the engine has at least one request in flight, and *idle* time, during which it has none. The busy component is governed by $W_{\text{busy}} \approx (G/\overline{R})\,\tau$ (Eq. 3), where $\overline{R}$ is the mean number of running requests, $G$ the generated tokens and $\tau$ the per-step cost. An admission policy moves $\overline{R}$, and on our configurations it moves little else—though §III gives two arms where the busy and idle components move in *opposite* directions, which is why the split is worth making at all. The difficulty is that $\tau$ is not constant in $\overline{R}$: across our runs $\tau_1$ spans 27.0–36.7 ms, and on the device-only family, raising the busy-time running count from 1.00 to 1.84 absorbs a third of what the extra concurrency buys.

The reason $\tau$ moves is recomputation, and whether recomputation happens is decided by the memory tier rather than by the policy. A session's prefix is worth reusing only while it is still resident, so prefix *reuse* is bounded by the workload—in our trace, 75.5% of the prompt tokens, a bound no configuration we ran reaches. How much of that bound is realized is where the tier enters. With the KV pool on the device alone, realization falls as concurrency rises—92.5% of the bound at $\overline{R}_{\text{busy}}$ 1.00 down to 36.4% at 1.84—and the shortfall appears as recomputation, 8.6 M to 20.7 M tokens. With a host tier in front of it the decline disappears: over the running counts our controllers occupy, the static points fall only from 92.4% to 90.2%, and the worst of the nineteen host-tier runs—the one that disables every throttle—still realizes 72.8%, where the nearest device-only point (the cap of four, at $\overline{R}_{\text{busy}}$ 1.65) manages 60.7%, while the *device* misses more. Two configurations at the same running count and the same pool occupancy realize 79.0% and 91.3% of the bound respectively (§III); what separates them is not how full the pool is but what happens when it overflows. That pair also differs in date and in GPU, so a capacity control that changes

the pool size alone on one card is queued behind the re-runs; §V reports it when it lands. That is the finding we would carry to another system first, and it is why this paper's answer to its own title is a property of the deployment rather than of the controller.

The idle component is structural rather than a control failure. A session in a tool wait issues no request, so when all live sessions happen to be waiting, the engine is genuinely empty. The stretches are short in the middle and long in the tail—mean 20.9 s—and the tail is what a scheduler feels: 69.9% of all idle *time* sits in stretches over 20 s, an order of magnitude above the 2 s control cycle. (The median is not a good summary here: it moves between 5 and 10 s depending on which runs are pooled, because one long run contributes a fifth of the stretches.) Crucially, the tool wait itself is much longer than the trace's median suggests: the median inter-round gap is 0.35 s, but the *mean* is 15.8–16.9 s with 22% of gaps over ten seconds. A controller that reads only the median concludes, as we initially did, that sessions never leave and therefore that there is no idle time to recover. There is.

### C. Contributions

- **We identify the memory tier as what decides how much of the achievable prefix reuse survives added concurrency, and therefore how much a controller has to gain by admitting more.** The two senses of "room" need separating, and §VI is what separates them: the tier bounds the *local* return on added concurrency (busy-time throughput), and it does not by itself decide whether a controller wins *end to end*, where idle time and the SLO axis also move. Our second workload is the demonstration—it has no host tier and its controller runs at low concurrency (1.22), the two conditions this section associates with room, and it still does not win (§VI). The claim here is the local one. The trace bounds reuse at 75.5% of its prompt tokens, and no configuration we ran reaches that bound—the best realizes 92.5% of it. What the tier decides is how much of the bound is realized as concurrency rises: device-only, realization falls from 92.5% to 36.4% between $\overline{R}_{\text{busy}}$ 1.00 and 1.84 and the shortfall appears as recomputation, 8.6 M to 20.7 M tokens; with a host tier it holds at 90.2–92.4% over 1.25 to 1.73, while the *device* misses more (§III). Two configurations at the same running count and the same pool occupancy realize 79.0% and 91.3% of the bound, so it is not how full the pool is but what happens when it overflows; that pair differs in date and GPU as well, so a capacity control that varies the pool alone on one card is queued; §V reports it when it lands. The per-step elasticity follows the same line: it passes one above $\overline{R}_{\text{busy}} \approx 1.5$ device-only and stays below one across the controllers' whole operating range with a host tier (§IV). Two measurements that do not share a code path—client-side step timing and engine-side token accounting—give the same answer. This is the finding we would carry to another system first, and it is what the title's question turns on.
- We test four routes by which an adaptive admission controller was expected to beat a tuned static cap on this workload, and we are precise about what each one rests on. Two are falsified by measurement against criteria we fixed before the runs—changing the accounting base (§V) and changing the growth reserve. One is a knob rather than a hypothesis, and we report it only because it slides the operating point along the front and adds no mechanism. The fourth, changing the execution granularity from session count to running-request count, is falsified by an argument about why the permit mechanism cannot express the control it would need; the one arm we got to run does not test it, because its gate bound in 0.6% of the cycles where a session was waiting. We do not call the last two of these falsifications of a stated hypothesis, because they were not.
- We give a two-part decomposition of the wall clock (§III) that locates the cancellation: admission policy moves the running-request count, and the per-step cost moves with it. On the device-only configurations we can account for that rise token for token—evicted prefixes are recomputed almost one for one. On host-tier configurations recomputation is flat and the account has to come from elsewhere; we report the part of that residual we were able to attribute, and the part we were not (§III, §IX-0e).
- We reduce the question to one measurable number, the elasticity of per-step cost with respect to the running count (§IV), and measure it at $0.38$ on the model-consistent step cost and $0.61$ on the per-request one, on the configurations whose mechanism we can account for token for token. We offer it as a diagnostic rather than a predictor: it says where on the curve a policy operates and how much the remaining headroom is worth, and §IV is explicit about the sense in which it is circular and the sense in which it is not.
- We characterize when adaptivity *could* have paid, and show that our workload is not that case: the same throttle configuration that is worth $+51\%$ at 25 sessions costs $-42\%$ at 200 (§VII). Put differently, no single scalar cap is optimal across both, which is the regime adaptive control exists for. On the low-load tier, where that argument bites hardest, the controller is *not* worse—same-generation measurements put it on the front, $+0.9$ pp above the line and inside the noise. On the high-load tier it is dominated. That asymmetry, not a blanket failure, is what this paper measures.
- We make the measurements reproducible: the front and each controller's signed distance are produced by one script from the raw per-step logs, the tables and the sub-thousand-step runs by two others, all named in §IX-0e. We report the six claims we had to retract along the way, with the checks that caught them.

### D. What we do not claim

We do not claim that admission control is useless in general. We claim it does not pay *here* and we show the frontier measurement that decides that; the diagnostic of §IV is a cheaper check an engineer can run before committing to a sweep, not a verdict on its own, and it can be taken on another workload before committing to a sweep. We also do not claim our hardware and model are representative; our results are from one 24 GB accelerator and one 7B model, and extending them is future work.

## II. Background and Setup

### A. Workload

We replay 200 coding-agent sessions drawn from the TRACELAB corpus [1]. Each session is a sequence of rounds; a round presents a prompt of median 23,187 tokens (p90 51,064; maximum 89,436, which alone is 88% of the KV pool), and between rounds the session waits on a tool before returning with a longer context. Arrivals are Poisson at $\lambda = 0.04\,\mathrm{s}^{-1}$ over an 80-minute window, giving 999 steps in total. Two sessions of maximum size cannot be co-resident: their combined contexts (89,436 and 87,821 tokens) exceed the pool by 75%.

The tool wait is the load's defining property and the one most easily misread. Its *median* is 0.35 s, which invites the conclusion that sessions never leave the engine. Its *mean* is 16.85 s measured and 15.82 s demanded, and the maximum is 300 s; 21.8% of the tool waits the trace specifies exceed ten seconds—18.0% of the 799 round transitions, 14.4% of its 999 steps—and 18.6% of the measured gaps do. We verified that our replay harness honors the trace faithfully: across 799 round transitions the measured gap is 16.85 s mean against a demanded 15.82 s. On a typical transition the harness adds nothing—the median of the per-transition measured/demanded ratios is 1.001—while the ratio of the two means is 1.065, the difference sitting in the tail. The corpus's own characterization makes the same point qualitatively—it reports "diverse and heavily-tailed tool calls" and calls these gaps "human-paced"—but gives no numeric mean [1], so the 16.85 s above is our measurement over the released trace, not a figure we quote from that paper.

### B. Engine and the binding constraint

We use SGLang v0.5.10 [2] with Qwen2.5-Coder-7B-Instruct [3] extended to 96K context, on one RTX 3090 (24 GB). KV is paged [4]; at `mem-fraction-static` 0.85 the pool is 101,432 tokens. Hierarchical caching (HiCache), with a host tier of twice the device pool, is enabled on the configurations marked *+HC* and on every Foyer run; the five plain static caps run without it. We keep both because the host tier turns out to change the mechanism (§III). Every run is guarded: if the pool does not come out at $101{,}432 \pm 20$—which happens when another tenant holds memory on the card—the run is aborted rather than recorded, because SGLang sizes the pool from free memory at startup and a shrunken pool silently produces incomparable numbers.

The relevant consequence is that the pool holds only about four median sessions. Concurrency is therefore capacity-bound, not scheduler-bound, and this is the premise on which everything below rests.

### C. The controller under test

Foyer is an admission controller that reads engine telemetry and issues *named* permits rather than a scalar cap: a session is admitted by appearing in a permit file, and the runner refuses to start a round for a session that is not listed. This design choice matters for the comparison, because it is what allows per-session actions at all—including the pause primitive we exploit in §V-H. It also means that a session, once admitted, holds its permit for its entire lifetime, so the controller never frees capacity by discarding a session's state—it admits fewer sessions instead. That is a statement about the *policy*, not about the memory: the engine's radix cache still evicts parked prefixes under pressure, and we measure that it does and that they are recomputed (§III). We return to this distinction in §IX, because it bounds how far our negative result generalises.

Each control cycle (2 s) the controller computes

$$\Pi \;=\; \max(\rho, \textstyle\sum_i c_i) \;+\; \sum_i g_i, \tag{1}$$

where $\rho$ is the engine's reported occupancy, $c_i$ the context length of live session $i$, and $g_i$ its forecast growth over the next `horizon` rounds. A session is admitted when $\Pi + n \le B$, with $n$ the candidate's estimated need and $B$ the budget. The three terms are respectively the measured physical occupancy, a reservation for sessions currently parked in a tool wait (whose KV is evictable and therefore absent from $\rho$), and a reservation for future growth.

### D. Baselines and the verdict criterion

Baselines are static caps: at most $N$ sessions resident, $N \in \{1, \ldots, 5\}$, with and without HiCache. We compare on two axes: end-to-end wall clock (time from the first submission to the last completion) and SLO attainment (fraction of requests whose time-to-first-token is under 10 s).

The verdict criterion deserves care, because "faster than cap-$N$" is not the question. The question is whether a policy reaches a point *above* the static front. A point that lies merely between two static caps is not a win: mixing cap-$N$ and cap-$M$ by time slice approximates any point on the line between them. We flag this segment argument as an argument, not a measurement—we did not run the time-sliced policies, and carrying queue state and cache contents across a switch could bend the line. It is the criterion we apply, and it is the stricter of the two we could have used—stricter than comparing each controller against the discrete static points alone, because the line sits above them. We state the direction plainly rather than claim it as a caution: this choice makes our negative result *easier* to reach, not harder. Our best arm is $0.55$ pp below the line and would be $+4.4$ pp against the nearest static anchor below it in wall clock (`cap3`+HiCache's repeat, 81.9% at 230.3 minutes; $+5.2$ would be the comparison against that anchor's first run at 81.1% and 236.4 minutes, which is the one

it displaced). The arm we repeated most tightly is a different one—`horizon`=1, at $0.93$ pp below the line and $+3.7$ pp against the same first run—and we name both because "best" depends on which of the two the reader means. So we take the upper-left envelope of the static points and report each controller's signed distance above or below it. We classify a signed distance of more than $+1.0$ pp as above the front, less than $-1.0$ pp as below it, and the band between the two as *on* it—a tie, not a win. The band is a convention we fixed before running anything, and we report it here because every verdict in this paper depends on it. Our later replicate measurements (§V) bracket it rather than justify it: the SLO spread of configurations we ran more than once ranges from 0.5 pp to 5.1 pp, so for some configurations the band is generous and for others it is far too narrow. We flag this as a weakness of the criterion, not a defence of it. One further property of the envelope deserves flagging, because three of the nine controller variants finish slower than 295 minutes. Its right-hand segment runs from `cap2`+HiCache at 295 minutes to `cap1` at 527, and `cap1` is the only anchor there that runs *without* HiCache. Those three—`hz2`, `breshz1` and `foyfix`—are therefore judged against an interpolation between one host-tier point and one device-only point. If the envelope is drawn from the host-tier anchors alone, that segment does not exist and those three cannot be adjudicated at all; drawn from device-only anchors alone, they sit further below. We report the conservative reading and note that the honest statement about those three is *below the front as we defined it*, not *below every static configuration*. A fourth arm, `bres`, sat in the same position in an earlier version of this paper; its re-measured value is 265.9 minutes, which puts it back inside the device-only range, so the class has three members and not four. Two of the three (`hz2`, `breshz1`) were also measured under host co-tenancy and are marked ‡ in Table I; whether they are slower than 295 minutes at all is one of the things the re-runs will settle. The front and every controller's signed distance are computed by one script from the raw per-step logs; the table generator computes the distance for the four runs that did not complete all 999 steps—both are named in §IX-0e—and that script's own first version was wrong: it extrapolated past the end of the envelope and declared the slowest static cap a winner. We fixed it and report the episode because it is the failure mode this criterion is most prone to.

## III. Decomposing the Wall Clock

### A. Two components

We split the wall clock at every sampled instant according to whether the engine has any request in flight:

$$W \;=\; W_{\text{busy}} \;+\; W_{\text{idle}}. \tag{2}$$

Fig. 2 shows the split for the twenty-eight runs of Table I's families that we have per-step logs for. Two things are immediately visible and both matter later.

First, idle time is not noise. Across the runs it spans 7.3% to 41.9% of the sampled span—7.3% for the fastest configuration, 36.3% for the slowest, and 41.9% for the one that idles most (all-throttles-off). The denominator here is the span the sampler observed, which runs about six minutes longer than the request wall clock of Table I, so the fractions against that wall clock are 7.6% and 36.7%—and it is not spread evenly. Over 1,844 stretches the mean is 20.9 s and 17.5% of them exceed 20 s, accounting for 69.9% of all idle *time* (Fig. 3). The tail, not the middle, is what a scheduler feels, and it sits at the scale of the 15.8–16.9 s mean tool wait (demanded and measured; `toolwait.csv`). We report the mean and the share rather than the median here because the median is not robust to which runs enter the pool: it is 10.0 s without `cap1` and 5.0–5.5 s with it, since that one run contributes 411 of the 1,844 stretches and is itself the longest. The mean (18.8–20.9 s) and the over-20 s share of idle time (64.6–70.1%) move by little across the four poolings we report in `idle_pooled.csv`, so the claim that the tail dominates does not depend on the choice. The 90th percentile moves with the median (25.1–30.1 s) and we give it as a range for the same reason.

Second, the two components move independently and in opposite directions, so neither is a proxy for the wall clock. The no-control baseline (§V) cuts idle time by 13.8 minutes against the base configuration—it admits everything, so nothing queues—and finishes only 2.7 minutes sooner, because its busy time grows by 11.1 minutes; the trade costs it the SLO column outright, 89.8% down to 3.0%. The arm that disables every throttle moves the other way, and further: idle *rises* by 129 minutes while busy *falls* by 5.4, so the 124 minutes it adds to the wall clock are almost entirely waiting, bought with 43.7 pp of SLO. Any account that treats "more concurrency" as one thing will get this wrong in one direction or the other.

Neither example is one of the damaged runs of §IX-0e; both are clean in the co-tenancy audit. Neither is a member of the controller family whose clean step-cost range that section quotes, either—the baseline is not a controller at all, and its 39.3 ms comes with the highest running count we measure (1.91), while the throttle-free arm runs a different controller. The charge-free base arm is a third case, whose split we give in §V with the re-measured numbers.

### B. What determines busy time

During busy time the engine executes steps; each step produces one token per running request. Hence

$$W_{\text{busy}} \;\approx\; \frac{G}{\overline{R}}\,\tau, \tag{3}$$

with $G$ the total generated tokens (fixed by the trace at 706,010) and $\tau$ the per-step cost. Across our runs $\tau$ is not a fit parameter we tuned; we obtain it two independent ways, and reporting both is what shows they are measuring different things rather than confirming each other.

*a) Method 1 (derived).:* $\tau_1 = W_{\text{busy}} \cdot \overline{R}/G$. Two choices in that expression are easy to get wrong and both matter. The time term must be the *busy* component, not the wall clock—using the wall clock charges idle time to the steps. And $\overline{R}$ must be the mean running count *during busy time*, not the

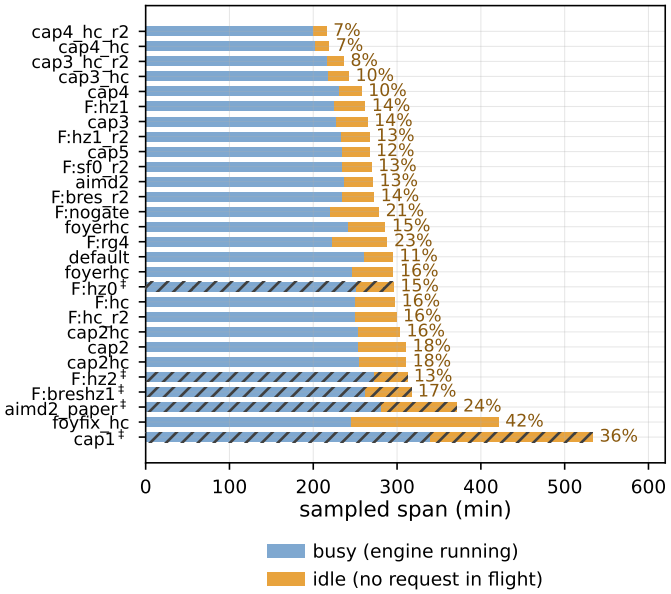



Fig. 2. Wall clock split into busy and idle for the twenty-eight runs of §III and §V, ordered by total duration. Idle is 7.3% for the fastest configuration and 36.3% for the slowest. Note the no-control baseline: it sits sixth of twenty-eight on idle share (11.4%) and yet sixteenth on duration, because its busy time grows with everything it admits. Five bars are hatched—the same ‡ flag as in Table I: three of the five have a measurably inflated step cost and two are flagged windows that left no measurable trace (§IX-0e).

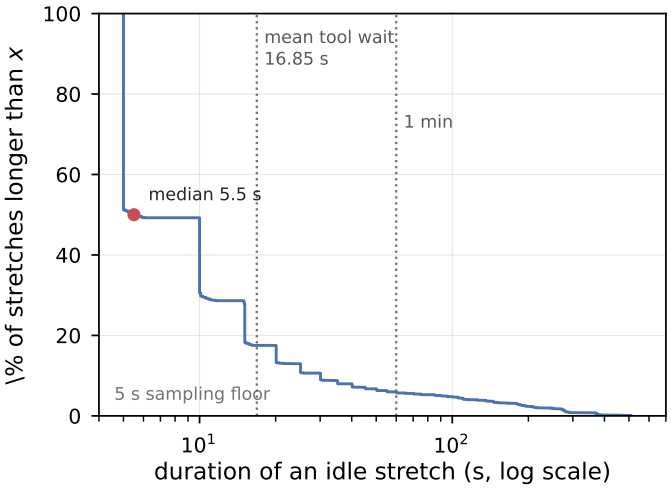



Fig. 3. Survival function of idle-stretch duration, pooled over all completed runs (1,844 stretches). Most stretches are short—the median is 5–10 s depending on which runs enter the pool, which is why we do not lead with it—but the minutes are not: 69.9% of all idle *time* sits in stretches longer than 20 s (64.6–70.1% across the four poolings in `idle_pooled.csv`), and the mean stretch is 20.9 s, the same order as the 15.8–16.9 s mean tool wait. The cliff is the 5 s sampling floor.

mean over the whole run: the whole-run mean includes the idle samples, where the count is zero, so it equals $\overline{R}_{\text{busy}}\times$ (busy fraction). Using it makes $\tau_1$ too low by exactly the busy fraction—so the estimate is short by the *idle* fraction, 36.3% for `cap1`, whose busy share is 63.7%. We report both columns in the data, and use the busy-conditional mean throughout; the whole-run mean is what an earlier revision of this paper used, and it made the two methods agree far better than they do.

*b) Method 2 (independent).:* For each step record we compute the per-request decode interval, $(\text{total_duration} - \text{first_token})/(\text{output_len} - 1)$, and take the median over steps with output_len $\geq 32$.

The two do not agree exactly, and the disagreement is informative rather than embarrassing. $\tau_1$ exceeds $\tau_2$ on every run we accept, by 7.7–39.7% (`decomp.csv`, column `excess_pct`), which in absolute terms is 2.6–8.2 ms of a 27–37 ms step. (The one run we exclude from the reuse accounting, the `horizon`=1 repeat, reads 66.3% here against its original's 18.5%. The anomaly sits in the engine's token counters and in everything derived from them, so that is the scope of the exclusion: this run's $\tau$ columns, its reuse and recomputation are unusable, while its wall clock and SLO—which come from the client—are ones we use, and it appears in Table I and in the repeat spread of §V on those two columns alone.) The gap is *consistent with* engine steps that are prefill and therefore emit no output token—we can bound it, not identify it, because subtracting a median from a mean does not isolate a component. All we can form from these two is their ratio: the excess $\tau_1/\tau_2 - 1$ runs 17.7–39.7% across the five device-only configurations. Two things follow, and we separate them. First, it is a ratio of *times*, not of steps, and we have no measurement of the fraction of step *counts* that are prefill—chunked prefill makes those two different numbers. Second, even as a time ratio it is the gap between two measurements, not a decomposition of one: it is consistent with prefill and non-decode work filling it, but we cannot rule out that some of the difference is the two estimators disagreeing for other reasons. So the two are not two estimates of one number; they are measurements of two different things, and we do not size a component with their difference. Fig. 4 shows both, run by run. Method 2 remains the one that does not depend on the quantity it is used to explain, and it is the one we quote when the question is the cost of a decode step. The engine's own latency counters (`itl_p50`, `e2e_p50`, `ttft_p50`) are exported as zero by SGLang v0.5.10 in our configuration, so no directly measured step time exists in our data; instrumenting this is future work.

### C. The per-step cost is not constant in concurrency

Equation 3 would be uninteresting if $\tau$ were a constant. It is not. Across the thirteen runs we fitted—the five device-only caps, the three host-tier caps that span the same range, and the five controller arms we report individually—$\tau_1$ ranges from 27.0 to 36.7 ms and $\tau_2$ from 20.6 to 33.1 ms, and both rise with the running count: fitting Eq. 4 to the five device-only configurations gives $\varepsilon = 0.38$ for $\tau_1$ and 0.61 for $\tau_2$ ($R^2 = 0.89$ for the latter, Fig. 5).

The relation is a trend, not a law. Two host-tier configurations with similar running counts behave differently: `cap4_hc` runs 1.73 requests at 29.8 ms while `F:hz0` runs 1.66 at 35.7 ms. Running count sets a floor and a trend, not a

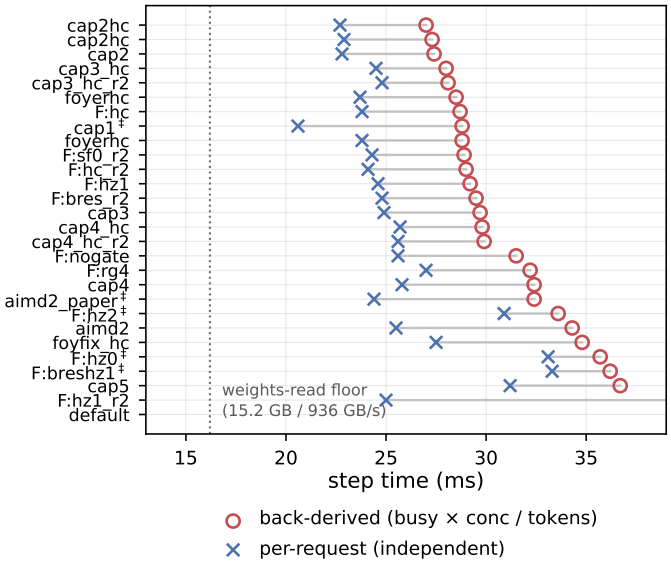



Fig. 4. Per-step cost by two independent measurements, one run per row: circles are model-consistent (all engine steps), crosses are the per-request decode interval. They do not agree—$\tau_1$ exceeds $\tau_2$ by 8–40%— and the gap is consistent with the prefill share, not measurement error; §III bounds that share rather than identifying it. The dotted line is the cost of reading the weights once; every run sits above it. Five rows are marked ‡—measured while other jobs shared the host (§IX-0e)—and three of the five have a measurably inflated step cost, which is why the marked rows cluster at the high end; the other two carry the flag without measurable damage.

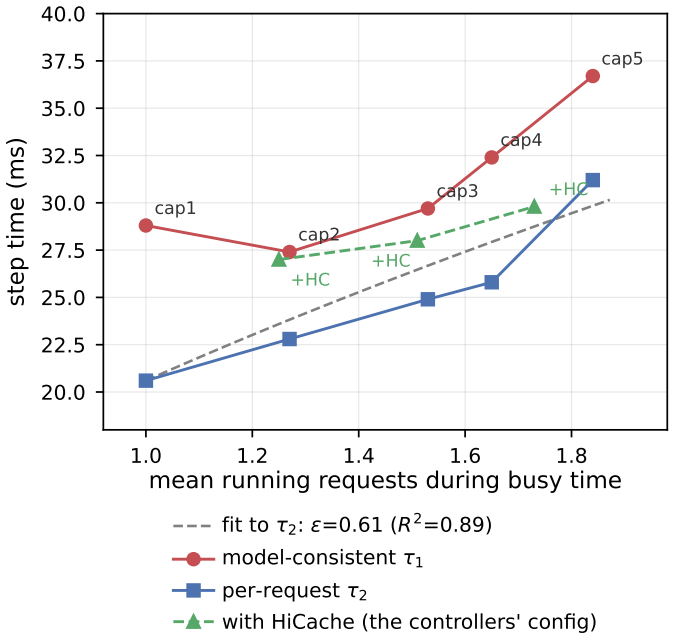



Fig. 5. Per-step cost against the running count during busy time. Solid circles are $\tau_1$ and squares $\tau_2$ for the five device-only configurations; $\tau_1$ is model-consistent (it averages over all engine steps, including prefill that emits no output token), $\tau_2$ is the per-request decode interval, and $\tau_1$ exceeds $\tau_2$ by 8–40%. The green triangles are $\tau_1$ for the three HiCache configurations—the ones the controllers run—and they are drawn against the same axes to make §IV's point visible: over $\overline{R}_{\text{busy}}$ 1.25 to 1.73, which covers the whole controller operating range, the host-tier curve rises 27.0 to 29.8 ms while the device-only curve over a comparable span rises 27.4 to 36.7. We plot $\tau_1$ only for the HiCache points to keep the figure readable. The fitted exponent is the elasticity $\varepsilon$ of Eq. 4; it is fit to the device-only points alone.

value—and on the host-tier family, as we show below, it does not set the value at all.

This is the crux of the paper, and we state it at the strength the arithmetic supports. Hold $\tau_1$ at its value for `cap1`, 28.8 ms, and busy time falls as $1/\overline{R}$: raising the running count from 1.00 to 1.84 should have cut busy time by a factor of 1.84, from 340 minutes to 184. The measured curve flattens and lands at 234. The rise in $\tau$ takes back 50 of the 156 minutes that the extra concurrency would otherwise have bought—*a third* of the gain, not all of it and, as an earlier revision of this paper claimed, not more than half. The other two thirds are real: raising concurrency does make the engine faster. What it does not do is make it faster in proportion, and the shortfall compounds with every controller that tries to exploit the difference.

*a) What the engine is doing when it is not decoding.:* SGLang exposes two counters that answer this directly: tokens evicted from the radix cache, and prompt tokens that had to be prefilled rather than reused. Across the five static caps *without* HiCache the two track each other almost exactly. `cap1` evicts 9.2 M tokens and prefills 8.6 M uncached; `cap5` evicts 21.3 M and prefills 20.7 M; the gap is 0.6–0.7 M on every one of the five. An evicted prefix is recomputed, close to one token for one, and that recomputation is chunked into decode steps. Those same five runs give recomputation rising with $\tau$ across four of the five points: 9.2 M uncached tokens at 27.4 ms, 11.5 at 29.7, 15.4 at 32.4, 20.7 at 36.7. The exception is the first point, `cap1`, which recomputes least (8.6 M) and still costs 28.8 ms—more than `cap2` does for more recomputation. We report the ordering we have rather than rounding it to a monotone one: it is a mechanism consistent with the $\tau$ trend on four of the five device-only configurations, and one point it does not order. It is five points on five different policies, not an intervention: we did not hold concurrency fixed and vary eviction, so we call it consistent rather than demonstrated.

*b) Where the mechanism stops.:* With HiCache enabled the correspondence breaks—and it breaks in a way that confirms the mechanism rather than undermining it. The device cache misses as much as before: `cap4_hc` misses 22.4 M tokens on the device, more than the device-only `cap4` misses. But only 9.1 M of that is recomputed; the other 13.3 M is served from the host tier. Across the nineteen host-tier runs the host serves 9.5–19.3 M tokens each, and across the five device-only runs it serves almost nothing—0.006% to 0.25% of the device miss, the rest of which is recomputed on all five. Recomputation is still what the device cannot absorb; the host tier is a second absorber in front of it.

### D. The tier, not the policy, decides whether reuse survives

The device-miss column hides the quantity that matters. Define *reuse* as the prompt tokens served from anywhere in the memory hierarchy—device radix cache plus host tier—so

that reuse + recomputed = prompt, with the prompt fixed at 28.5 M by the trace.

The bound on reuse has to be defined independently of what we observed, or the argument is circular. We take it from the trace: summing the planned reusable prefix (`prefix_len`) over all 999 rounds of `replay_pois200_l04.csv` gives 21,540,371 tokens against a prompt total of 28,526,871, so no policy can reuse more than 75.5% of the prompt. That is a property of the workload. It is also a bound rather than a description: *no configuration we ran reaches it*, and the best realizes 92.5%, so the gap between "planned" and "achieved" exists even in the easiest case. What varies across configurations is how much of the bound is realized, and that is where the tier enters.

| | | | | | | |
|---|---|---|---|---|---|---|
| device-only | $\overline{R}_{\text{busy}}$ | 1.00 | 1.27 | 1.53 | 1.65 | 1.84 |
| | realized (% of bound) | 92.5 | 89.7 | 79.0 | 60.7 | 36.4 |
| | recomputed (M) | 8.6 | 9.2 | 11.5 | 15.4 | 20.7 |
| host tier | $\overline{R}_{\text{busy}}$ | 1.25 | 1.51 | 1.73 | | |
| | realized (% of bound) | 92.4 | 91.3 | 90.2 | | |
| | recomputed (M) | 8.6 | 8.9 | 9.1 | | |

Device-only, realization falls from 92.5% to 36.4% and the shortfall is recomputation: 8.6 M to 20.7 M tokens. With a host tier it holds at 90.2–92.4% on the static points that span the same range, and at 72.8–92.5% across the nineteen host-tier runs that completed this trace—the same trace, HiCache on, at least 995 of its 999 steps—including every controller arm; the lowest is the arm that disables every throttle and therefore evicts the most (21.1 M tokens, the largest in the table). The *device* meanwhile misses more than the device-only family's at every point (21.6–27.9 M against 8.6–20.7 M), because the host serves 9.5–19.3 M in front of the recomputation instead of letting the device keep it—so a reader watching only the device hit rate would conclude the host tier made things worse. That rule on its own selects *twenty*, and we name the four runs it is not meant to catch, because a reader scanning the logs will find them: two that stopped at 763 and 770 steps (they realize 70.7% and 67.1%, and we exclude them for not finishing rather than for their values), the variable-load family (a different trace, and short of the step count as well), and one whose engine prompt-token counter reads $1.41\times$ the trace plan with zero retractions and no client-side retries, which we could not reconcile. (Two denominators appear here and we name them: the 75.5% is the trace's bound as a share of the prompt, and 92.5% is the best observed reuse as a share of *that bound*—`cap1`, the run that realizes it, reuses 69.9% of its prompt. `reuse.csv` carries `reuse_frac` and `realized_frac` side by side so the two cannot be confused.)

This is the same collapse §VIII reports under a different normalization. That section quotes the round-1-and-later prefix hit rate—steps restricted to rounds $\geq 1$, unweighted mean over steps—which falls from 80.8% to 32.8% across the device-only caps and holds at 80.7% to 78.7% with a host tier. The two conventions differ in two ways, not one: which steps are in the set (round $\geq 1$ against all rounds) and how steps are combined (unweighted mean against token-weighted sum). We use the token-weighted all-steps convention here because reuse and recomputation must add to the prompt, and we name the other whenever we quote it. Both are *realized* quantities that vary with the policy; neither is the trace's planned rate, which is fixed at 75.5% and appears in this section only as the bound.

Device occupancy does not explain the split. `cap3_fixed` and `cap3_hc` run at nearly the same mean device occupancy (55.8% and 55.5%) and nearly the same running count (1.53 and 1.51), and they realize 79.0% and 91.3% of the bound respectively. The host-tier side of that comparison is not one run: `cap3`+HiCache has a repeat which sits at the same occupancy (57.4%) and the same count (1.53) and realizes 91.1%, so the 12-point gap is not one lucky run on the host-tier side. The device-only side is one run, and we say so rather than implying a distribution. What differs is not how full the device pool is but what happens when it overflows: with no host tier an evicted prefix is gone and must be recomputed; with one it is demoted and read back. The pool is the binding resource; the tier sets the pool; and in these two runs, which differ in the tier and in nothing we can see on the engine side, the tier is what separates paying for added concurrency in recomputation from not paying for it. The caveat is that this pair is two runs on two different days and two different GPUs, not a paired experiment; §IX-0e states what we did to rule out co-tenancy. A capacity-controlled version—same card, same trace, only the pool size changed—is queued behind the re-runs and we report it in §V when it lands; we do not use it here.

*a) The second trace agrees on direction and not on level.:* We ran the same accounting on a second workload, reported in full in §VI—117 real Claude agent sessions, 390 rounds, the same pool and the same engine—and it reproduces the direction: its realization of *its own* bound falls from 74.8% to 57.5% as the running count rises from 1.08 to 1.50, and over that span the whole-range per-unit slope is $-40.9$ points, against the device-only family's $-66.8$ over $\overline{R}_{\text{busy}}$ 1.00–1.84 (`slope.csv`). What it does not reproduce is the level, and this is worth stating because it bounds what we can claim. The lightest cap on that trace (`realhr_cap2`) runs at a *lower* median device occupancy than the lightest on this one (15.8% against 22.4%, `cap1`) and realizes *less* (74.8% against 92.5%), so realization is not a function of occupancy alone—and a statement of the form "the pool collapses above $x$% occupancy" is wrong on our own data. Occupancy is instantaneous while reuse depends on history; a session's prefix must survive from one round to the next, and how much pressure that puts on the pool is not visible in what the pool holds at an instant. We do not have the measurement that would separate those, which is why the controlled experiment now queued for §V—it varies the pool alone, on one card and one trace—is the one that will settle it, and why we do not lean on the cross-trace comparison here. The claim this section supports is the narrow one: within a configuration, the host tier preserves realization at a matched running count and a matched occupancy, and without it realization falls.

That gives the criterion of §IV a mechanism rather than

a correlation, and it suggests a prediction we are testing: if the tier acts by multiplying usable capacity (the host pool is twice the device pool, so three times in total), then shrinking the device pool alone should move the collapse point even with no host tier involved. We do not treat the arithmetic as a result—it assumes KV demand grows linearly in the running count, which heterogeneous sessions need not satisfy—and the queued experiment measures the collapse point directly rather than predicting it, and §V reports it once it has run.

*b) What is left unexplained.:* Less than an earlier draft of this paragraph claimed. Across the nineteen host-tier runs recomputation spans 8.6–12.8 M tokens and realization 72.8–92.5%, and $\tau_1$ spans 27.0–36.2 ms. Three of those runs (`F:hz2` 33.6, `F:hz0` 35.7, `F:breshz1` 36.2 ms) were measured while three other engine instances shared the host, which inflates per-step cost by 24–41% and leaves the volume columns untouched (§IX-0e); they are being re-measured and we do not use their timing. A fourth was re-measured clean and is in the set above with its new value. Across the remaining sixteen the $\tau_1$ spread does not sit unexplained at all: it tracks their recomputation closely (a straight line through the sixteen leaves every run within 1.5 ms, $r = +0.92$), which is the mechanism of this section holding inside the host tier rather than a residual. We report the association and not the line: the coefficients are fitted to one configuration family and we have no derivation for them. We note that the negative result does not rest on it either way: the cancellation is measured directly, not inferred from the mechanism.

One caution about the strength of the account above, since the arithmetic invites overreading. reuse + recomputed = prompt is a definition, not a finding, so "the shortfall reappears as recomputation" follows from how we partitioned the prompt and cannot by itself establish the mechanism. What is independent of the partition is the pair of engine counters: evicted tokens and uncached prefill tokens are separate SGLang counters, and on the device-only runs they track within 0.6–0.7 M on all five—which is the observation that licenses reading eviction as the cause of recomputation. With a host tier the two come apart by design, and the 1.3–8.6 M gap between them is an upper bound on what the host absorbed out of the evicted pool: those tokens were evicted and not recomputed, which is consistent with the host having served them and equally consistent with their never having been asked for again. We can bound it, not identify it (`eviction.csv`); the host serves 9.5–19.3 M in total, and by the counter definitions those are tokens the *device* did not have when the session came back—it had evicted them—and which were not recomputed.

*c) A refutation we had to withdraw.:* An earlier revision withdrew this mechanism outright, on two checks that were both wrong. The first was that total prefilled tokens equalled the trace's planned prefill to a ratio of 1.00, so no token could have been computed twice. `prompt_tokens_total` counts the full submitted prompt *including* its cached portion, so it equals the sum of prompt lengths by construction and cannot detect recomputation; the quantity that can is the uncached count used above, which varies by 2.4× across these runs. The second was that SGLang reported no retractions. Radix-cache eviction is not a retraction—retractions count running requests preempted mid-flight, a different event with its own counter. We record the episode because it is the same failure mode as the segment argument of §II: a check that looks like a measurement but is an identity.

### *E. Why idle time exists, and why recovering it does not pay*

The length of an individual idle stretch is a property of the workload, not of the controller. A session in a tool wait holds its permit and its KV but issues no request; with only about two live sessions, the probability that all of them are simultaneously waiting is substantial. The direct evidence is the duration of the stretches themselves: they are of the same order as the tool waits, and an order of magnitude longer than the 2 s control cycle (Fig. 3). No controller can shorten a gap that the tool set.

How much idle a *run* accumulates is a different matter, and there the controller is not off the hook. Idle accumulates when every live session happens to be parked at once, and how many sessions are live is precisely what an admission policy sets. Two configurations with the same mean running count make this visible: `horizon`=1 (1.52) and `cap3`+HiCache (1.51) idle 36.7 and 24.1 minutes respectively—a 52% difference at matched concurrency—and 12.6 of the 19.4-minute wall-clock gap between them is that idle difference rather than step cost. So the gap durations are the workload's, and the number of gaps is partly the policy's.

We note one claim we had to drop here. An earlier draft reported that idle fraction falls monotonically with mean concurrency. With the full data it does not: on the busy-conditional measure this section uses throughout, `F:hz0` at $\overline{R}_{\text{busy}}$ 1.67 idles 15.2% while `cap4` at 1.65 idles 10.4%—more concurrency and *more* idle, the reverse of the reported trend. (The two arms read 1.41 and 1.48 on the all-samples mean, where the ordering happens to run the other way; `arm_running.csv` carries both conventions side by side, and we name the one each claim uses.) The weaker statement that survives is the one we use—the stretches are structural and long.

The obvious remedy—keep more sessions live so that someone is always ready—is what `bres` does, and the clean measurement of it says the remedy works and does not pay. `bres` cuts idle time by 9.5 minutes against its own baseline, and its busy time falls as well, by 15.5—so the arm simply finishes 25.4 minutes earlier. What it does not do is improve the criterion: its SLO falls 3.9 pp and its distance below the static front is unchanged (§V). Recovering idle time is not the same act as buying a better operating point, and here the one does not produce the other. We are reporting a correction in this paragraph, not just a refinement: an earlier draft had idle falling while busy rose, leaving the run slower, and that shape came from a measurement we have since found was taken under host co-tenancy (§IX-0e). Idle is recoverable; it is recoverable at a price that does not move the verdict. The structural remedy would be to make the engine do work during

a wait. For the session that is waiting there is none—its next round does not exist until its tool returns—but for the *pool* there is: the admission queue holds a median of 66 sessions waiting to get in, and a policy could run one of those instead of holding a permit for a parked one. §V measures what happens when it does: the charge-free base arm cuts total idle from 47.4 to 37.9 minutes, and it does not move the verdict. Idle is recoverable in part; recovering it is not what pays.

## IV. A Criterion for When Admission Control Can Pay

The decomposition turns the question into one number. Throughput during busy time is $\overline{R}/\tau$, so raising concurrency by a factor $f$ buys a factor $f^{1-\epsilon}$, where

$$\epsilon \;=\; \frac{\mathrm{d}\ln\tau}{\mathrm{d}\ln\overline{R}} \tag{4}$$

is the elasticity of per-step cost with respect to the running count. At $\epsilon = 0$ the step cost is flat, concurrency converts one for one into throughput, and an admission controller has room to work. At $\epsilon = 1$ the two cancel exactly and it has none. Everything in between is a quantitative statement about how much room.

On the five device-only configurations, where §III establishes that the step cost is driven by recomputation, $\overline{R}_{\text{busy}}$ runs from 1.00 to 1.84. A least-squares fit of Eq. 4 over those five points gives $\epsilon = 0.61$ on the per-request measurement ($R^2 = 0.89$), or $0.38$ on the model-consistent one (Fig. 5).

That fit is a straight line through a curve, and the curve is what matters. The segment elasticities are $\epsilon = -0.21$ from `cap1` to `cap2`, $+0.43$ from `cap2` to `cap3`, $+1.15$ from `cap3` to `cap4`, and $+1.14$ from `cap4` to `cap5`. Below $\overline{R}_{\text{busy}} \approx 1.5$ added concurrency still buys something; above it, $\epsilon$ passes one, which means busy-time throughput stops rising with the running count—the engine spends *more* busy time per generated token. That is not the same as the run getting slower end to end: `cap4` is 8.8 minutes *faster* than `cap3` in total wall clock even though its busy time is longer, because it collects less idle. The single number $0.38$ averages those regimes and describes none of them.

The operating region is the one that matters, and it is not the region the controllers run in. All nine controller arms run with HiCache enabled, which the five device-only caps above do not. Measuring the same elasticity on the three HiCache configurations we have gives $\epsilon = +0.19$ from `cap2`+HC to `cap3`+HC and $+0.46$ from `cap3`+HC to `cap4`+HC, over $\overline{R}_{\text{busy}}$ 1.25 to 1.73. That range covers the whole controller operating region (1.35 to 1.69), and over it $\epsilon$ never approaches one. At matched running counts the HiCache caps pay 1.7 ms less per step at $\overline{R}_{\text{busy}} \approx 1.5$.

This contradicts what an earlier draft of this paper claimed, and it changes what the criterion is for. The cliff in the device-only curve is a recomputation effect: §III measures that those runs recompute what they evict, while under HiCache the host tier absorbs it. The cliff does not survive on the configuration our controllers actually use. So the criterion does *not* explain why they lose—on their own configuration they were not in a region where added concurrency was self-defeating, and the engine had room they did not take. What the measurement does is rule one explanation out. The failure is not that this workload denies admission control its lever; it is that our controllers do not pull the lever they have, which is what §V-B and the attribution of §V-C measure directly.

We are explicit about what this does and does not establish. The HiCache curve rests on three configurations and two segments, one run each, with no error bars, and it stops at 1.73—we cannot see whether a cliff appears beyond the controllers' region. A $\pm 10\%$ perturbation of $\tau$ puts the segment elasticities' bounds across one even on the device-only curve, so the cliff is a description of the fitted points rather than a threshold we have located precisely. And it is measured on different policies rather than by an intervention, so it is a cross-policy association—the two configurations the tier comparison rests on were run a day apart on different cards, which §III states, and a same-card pairing is queued. On our data it is also computed from the same wall clock the model then explains—$\tau_1$ is derived from $W_{\text{busy}}$, so using it to predict $W_{\text{busy}}$ restates rather than tests—and its points are different policies, so the slope is a cross-policy association, not a transportable causal derivative. $\tau_2$ is genuinely independent of the wall clock but measures only decode, so it cannot be dropped into a throughput equation that includes prefill.

We tested whether the curve transfers to a second workload. On a real agent trace (117 sessions, median inter-round waits of 20.2 s against 0.34 s on the synthetic trace, `toolwait.csv`, six static caps and one controller arm) the model-consistent elasticity is negative over the first two segments—$-2.40$ then $-0.28$, at $\overline{R}_{\text{busy}}$ 1.08 to 1.29—and then turns positive above 1.29 ($+1.0$, $+0.18$, $+0.87$ over the three segments that reach 1.50). The $\varepsilon > 1$ regime is touched and not held there: the first of those three segments is the only one at or above one, and a $\pm 10\%$ perturbation of $\tau_1$ carries every one of their bounds across both zero and one. The main trace, for contrast, crosses one at 1.53 with $+1.15$ and stays above it. So the direction transfers and the location does not, and on this trace we can say no more than that added concurrency stopped paying. The per-request measure is no steadier on that trace ($+0.41$, $+0.48$, $+0.45$, $-1.25$, $+0.73$). The controller arm there runs at 1.22, below every one of the nine arms above and below where the branch turns expensive on either trace, and still does not beat the static front at the threshold this paper uses—though at a stricter one it does (§VI). We report this as a boundary: the criterion identifies a branch, and where we could check it elsewhere the sign of the branch transferred while its location did not; it never explained the failure. It is not, as an earlier draft claimed, something that decides the question in minutes before building anything.

## V. Results: Four Routes, One Outcome

### A. The frontier

Fig. 1 places every completed run on the (wall clock, SLO) plane and draws the upper-left envelope of the static points.

Table I gives the numbers and each controller's signed distance from that envelope.

The result is unambiguous for the arms we can compare, and we say which those are before we lean on it. Of the nine controller variants, six are clean measurements; of those six, *none* is above the envelope—not one, on any run, and the pair we repeated most tightly says so at a resolution the others do not. The closest arm of all is the one with single-flight admission disabled, at $0.55$ percentage points *below* the line—inside our $\pm 1.0$ pp tie band, so we record it as *on* the line rather than below it, but on the below side of it.

The six, and why not the other three: those carry the co-tenancy flag of Table I, their wall clock, SLO and $\Delta$ are not comparable with the rest, and their verdict is *provisional*. The direction of the correction is known and it runs our way—co-tenancy inflates the step cost, which pushes an arm *down*, so their measured distances overstate how far below the line they are; the question is only by how much. A fourth variant carried the flag and has been re-measured (§IX-0e): it moved from $7.81$ pp below the front to $1.33$, still below and still outside the tie band. That 6.5 pp is the largest correction the flag has produced so far, and it is a warning about extrapolating from it rather than a reassurance: applied unchanged to the other three as they stand, it would take `horizon`$= 2$ from $-4.81$ to $+1.7$—above the line—while leaving the other two below. One re-measurement is one data point, and the three differ in how long they overlapped and in what else was running, so we are not predicting that. We are saying that the correction *could* cross the line, and that we cannot rule it out from what we have. If one of them does, the headline weakens to "six of nine, one win", and this paper will say so.

The base configuration is the one whose repeats we can trust, and they are tight: four runs spanning 0.5 pp of SLO (89.5% mean; 279.8, 289.6, 291.3 and 293.2 minutes at 89.5%, 89.3%, 89.8% and 89.6%—the matched-pair experiment ran the same configuration twice more, as `pairA_foyerhc` and `pairB_foyerhc`), against the 3.3 pp that separates it from the single-flight-off arm. So that difference is several times the noise of its own configuration rather than inside it, and it is a trade rather than a win: 28 minutes of wall clock bought with 3.5 pp of SLO, moving the arm onto the tie band from below it. Not one configuration whose measurements are clean gets above the line, and the tightest pair we have is what makes us willing to say the near misses are misses. A fifth pair, two runs of `cap3`+HiCache, *is* used, and an earlier draft of this paper wrongly set it aside saying one of them exceeded its own cap. That was a different run: the over-cap one is the device-only `cap3` repeat (peak in-flight five against a cap of three; at our 300-point in-flight sample grid, 126 samples sit over the cap, a count that scales with the grid rather than with the run), while both HiCache runs peak at three with none. The repeat is also both faster and higher on SLO (230.3 minutes, 81.9% against 236.4 and 81.1%), so by the same rule that displaced `cap4`+HiCache's first run it displaces this one—and moving that anchor is what puts our best arm below the line rather than on it.[1]

The anchor repeats moved the frontier twice and we report both moves. The leftmost anchor's second run came out both faster (209.1 against 213.0 minutes) and higher on SLO (68.5 against 63.4), displacing the first; no controller verdict moves, because every controller finishes to the right of the next anchor. The middle anchor's repeat moved it the other way for us: it is also faster and higher, and taking it moves our best arm from *on* the line to *on the line, below it*. We report every repeat we ran rather than keeping the better of each pair, because a frontier whose anchors are single measurements is the weakness this whole subsection is about—and here it is the weakness biting.

The other end of the front deserves its own sentence, because the rest of this paper is negative and a reader could close it thinking the answer is to do nothing. The run with *no* admission control—every session admitted the moment it arrives—finishes in 288.6 minutes at 3.0% SLO, $87.6$ pp below the envelope and the worst SLO we report. Its median time-to-first-token is 1,774 seconds, its prefix cache hit rate collapses from the planned 75.5% the runner prints in its end-of-run summary (the all-steps convention, a different one from the round-1+ numbers in §VIII—for the same trace `gen_nocontrol.py` reports 73.8% on that convention) to 1.6%, and its segment SLO is 30% in the first tenth of the run and exactly zero in each of the nine after that. Uncontrolled admission does not degrade gracefully; it destroys the cache and then serves almost nobody. The question this paper addresses is therefore not whether to admit selectively, but which selective rule to use.

Two features of the front are worth noting before the negative results. The envelope is steep between 209 and 230 minutes (0.63 pp per minute) and shallow thereafter (0.15 pp per minute), so the cost of buying SLO by retreating from the fast end is high. And $W$ is not monotone in concurrency: five sessions (260.9 min) is *slower* than four (249.9 min)—the same configuration family, no host cache on either side—because the per-step cost rises faster than the request count does. We return to this non-monotonicity in §III, where it turns out to be the first appearance of the mechanism that drives the whole frontier.

[1]The repeats are uneven, and we do not want the tightest pair to stand for all of them. Five configurations have been run more than once; their SLO spreads are 0.5, 0.8, 1.0, 3.9 and 5.1 pp (`replicates.csv`). The two widest belong to the two arms that decide something—`horizon`=1 and the leftmost anchor—so where a small margin matters our repeats are weakest, and where they are strongest (the base configuration and the `cap3`+HiCache anchor) the margin is not small. We report both ends rather than the average. There is also an asymmetry in how the two sides enter the comparison, and it runs against the controllers: the envelope takes each anchor's *better* run where a repeat exists, while each controller is a single run. The exposure is bounded, and by less than the widest anchor's spread suggests. Every controller sits to the right of `cap3`+HiCache's repeat, so the verdicts are set by the segment between that anchor and `cap2`+HiCache—and those two anchors' own repeats differ by 0.8 and 1.0 pp, below the $\pm 1.0$ pp band we judge in. The leftmost anchor's 5.1 pp spread cannot reach any controller's $\Delta$, because no controller finishes fast enough to be judged against that segment.

### B. The controllers leave memory unused

Before the routes, one measurement that reframes all of them. Our controller keeps every session's permit for its lifetime, so its pool occupancy is a policy output, not a constraint. It runs at a median of 49.2% of the pool on the base configuration, and the median number of sessions queueing for admission is 66. The configuration that finishes fastest—the repeat of that same static cap of four, at 209.1 minutes—runs at 71.9%, and its first run at 70.2%. Occupancy and running requests track each other reliably across the arms we have (the two columns in `occupancy.csv` rank the twenty-three arms we have almost identically, Spearman $\rho = 0.95$), but running count does *not* order wall clock: `horizon`=2 runs 1.25 requests against the base arm's 1.14 (the all-samples mean, the column `occupancy.csv` carries; §III uses the busy-conditional one) and finishes 15 minutes slower. That gap is what the next subsection is about.

The controller believes it is nearly full. Its own decision log records the projection it admits against—the largest of measured occupancy and the summed contexts of live sessions, plus a reserve for their growth—at a median of 89,073 tokens against a budget of 96,360, i.e. 7.6% free. Its measured occupancy at the same instants is 49,875 tokens: 48.2% of that budget, or 50.8% of the pool, is idle. The projection is 1.79× the truth on this configuration. Across the sixteen controller runs that finished on this trace and logged projections the ratio runs from 1.32 to 1.90 with a median of 1.78, and from 1.32 to 4.11 with the same median once the second trace's controller arm is included (`projection.csv` records each set, its size and its median; the second trace is the 4.11). So 1.79 is this arm's value, not a constant of the design, and we name the set because the range depends on it—over every finished run in that file, including the flash-crowd and 25-session families, it is 1.01 to 4.11 with a median of 1.58. (Runs still in flight are excluded from both figures: their controller logs are still growing, and one of them reads 1.95 today.) Medians here are the upper of the two middle values, the convention §V-C uses throughout. Two terms produce the gap: charging every live session its full context, when a parked session's KV is evictable and therefore absent from the engine's count, and the growth reserve, which alone is 26,762 tokens, 28% of the budget.

Two things follow, and we are careful about the second.

The first is that the controllers under-use the pool, and that this is a property of their estimates rather than of the hardware: the base configuration holds nearly half the pool idle (48.2% of its budget, 50.8% of the pool) with 66 sessions queued, because it believes it has 7.6% free when it has 48.2%. Any account of why it is slow that does not mention this is incomplete.

We then asked what actually stops it, by instrumenting the admission decision. The path is a conjunction: a session is authorized only if the SLO valve is open, the round gate is not full, and its projected footprint fits what remains of the budget. Over the base run's 8,196 blocked cycles—cycles with a session waiting and none admitted—the valve accounts for 6.8%, the round gate for none, and a projection already above budget for 38.4%.

The remaining 54.8% are cycles where the controller selected no candidate at all, and here we are explicit about the limit of what we can resolve. The controller is single-flight by default: it authorizes at most one session per two-second cycle and waits for that session to begin a round before authorizing the next. Whether that gate held or whether a candidate was tried and did not fit is *not distinguishable from our logs*—the controller does not record it—so we do not attribute the bucket to either. We bounded it by running that arm: the same configuration with single-flight disabled. Two things came out of it, and the second is the one that matters.

The bound first. With the gate off, the residual bucket—which can then only be *candidates that did not fit*, since the gate cannot be holding—falls from 54.8% of blocked cycles to 39.7%, against 44.0% that are over budget outright. So the single-flight path accounts for roughly fifteen points, and the headroom estimate remains the largest single cause. (Both arms' numbers are rows of `gate_attrib.csv`, which now includes the `sf=0` arm; an earlier revision quoted 39.9 and 43.5 for the second pair, values that appeared in no data file.) We report what did *not* move, because it is the check we wrote down for this probe before running it. The two aggregate indicators barely did: admitting cycles go from 2.35% to 2.44% and blocked cycles from 93.3% to 93.2%, so on those two the flag looks like it changed nothing. What moves is the composition *within* the blocked cycles—which is the quantity the bound above rests on—and it moves well outside the replicate spread, the repeat of the base arm reading 58.3% remainder against the original's 54.8%. So the probe supports the decomposition and not the claim that disabling the gate makes the controller admit appreciably more, which is what we say above it buys 28 minutes for 3.5 pp rather than a win.

The second thing is that disabling it does not make the controller win. `sf=0` finishes in 263.3 minutes at 86.3% SLO against the base run's 291.3 and 89.8%: it buys 28 minutes for 3.5 pp, and its signed distance from the front moves from $-1.24$ to $-0.55$—onto the tie band from below it, not above it. Its admission rate is essentially unchanged (2.44% of cycles admit, against 2.35%). So the gate was a real cost and removing it moves the arm along the front rather than past it, which is the same trade every other route in this section makes.

Across the eight Foyer variants that left controller logs with gate state—the set is named in `gate_attrib.csv`, and it is eight rather than nine because the ninth, `sf=0`, is the probe we report separately just above, and because `foyfix` and the two AIMD arms run a different controller and log no gate state—the projection is over budget in a median 32.2% of blocked cycles (19.5% to 53.6%) and the valve in a median 32.7% (6.8% to 67.9%). Both medians move with the set, which is why we name it: over every completed run in that file—forty-two of them—they are 30.6% and 5.3%. One convention note, because it moves these numbers: we take the median as

the upper of the two middle values when the count is even (`v[len(v)//2]`, which is what `gen_gate_attrib.py` computes and what the file's `eight_median` line records), not the mean of the two; on eight arms that difference is 0.4 pp on the projection and 6.7 pp on the valve. The round gate is off by default and binds in only two of the three arms that turn it on—`rg2` at 29.3% and `rg3` at 18.0% of blocked cycles, while `rg4`, the loosest of the three, stays at 0.6% like every arm that leaves it off—so the gate binds only where it is set tightly enough to. We also record what this accounting omits, because it changes the picture: the controller has a starvation guard that bypasses the budget entirely once the oldest waiter has waited 240 s, cooling down for 300 s. On the base run it accounts for 24.8% of all admissions (51 of 206), and while the projection is over budget it is the *only* way in—of 3,700 such cycles, nine admitted, and all nine were forced. It is therefore the guard, not the budget test, that sets the floor on how fast this controller can admit.

The second is that under-use does *not* order the outcomes, so it is not the whole explanation either. `horizon`=1 and `cap3`+HiCache sit at the same median occupancy (57.8% of the pool) and the same mean running count (1.52 against 1.51 on the busy-time measure of §III), and the cap is 19 minutes faster. `hz0` runs at 64.1%, higher than that cap, and is 54 minutes slower. Occupancy buys running requests reliably—`occupancy.csv`'s occupancy and running-count columns track each other across all twenty-three arms—but converting running requests into wall clock is not simply the per-step cost of §III. Decomposing that 19 minutes into its idle and busy components, 12.6 of them are idle time—65%—and only 6.8 are busy; the two runs' step costs differ by 1.2 ms (29.2 against 28.0), about 4%. A parked session is one the controller has authorized but not yet released, so the same policy that sets the running count also sets a large part of the idle time. Both effects are real; on this workload the unused-memory one is larger for the arms that finish worst, and the step-cost one is what separates the arms that finish close together.

### C. What the two mechanisms explain, arm by arm

We have offered two mechanisms, and the honest thing is to say what each one covers. For every controller arm we took the static cap closest to it in wall clock and compared their occupancy and their step cost. An arm is attributed to the unused-memory mechanism if it runs fewer than 92% of its comparison's requests, and to the step-cost mechanism if it runs a comparable number—within 8%—at a measurably higher cost (more than 8% higher $\tau_1$). Everything else is a residual. The rule reads the running count and $\tau_1$ only; it does not consult occupancy, so the "unused memory" label is licensed by the running count alone—though the three arms it lands on also sit well below their comparison's occupancy (54.5, 56.2 and 62.1 against 74.3%), which is why we let the name stand. The comparability clause is load-bearing and we state it because an earlier version of this table did not enforce it, which classified every arm that ran *more* requests than its comparison as step-cost purely because $\tau$ rises with the running count.

| Arm | Occ. | $\overline{R}$ | $\tau_1$ | Attribution |
|---|---|---|---|---|
| single-flight off | 54.5 | 1.455 | 28.9 | unused mem. |
| `horizon`=1 | 57.8 | 1.524 | 29.2 | residual |
| Foyer base | 49.2 | 1.353 | 28.7 | residual |
| `horizon`=2 | 54.7 | 1.439 | 33.6 | $\tau$ high, not matched‡ |
| `horizon`=0 | 64.1 | 1.666 | 35.7 | $\tau$ high, not matched‡ |
| `bres` | 56.2 | 1.481 | 29.5 | unused mem. |
| `breshz1` | 61.4 | 1.632 | 36.2 | $\tau$ high, not matched‡ |
| no gate, no resv. | 62.1 | 1.678 | 31.5 | unused mem. |
| running gate=4 | 62.3 | 1.689 | 32.2 | $\tau$ high, not matched |
| all throttles off | 39.2 | 1.675 | 34.8 | $\tau$ high, not matched |
| AIMD, retuned | 66.6 | 1.706 | 34.3 | residual |
| AIMD, paper params | 40.3 | 1.357 | 32.4 | step cost‡ |

‡ marks runs measured while three or more other engine instances shared the host for at least an hour (§IX-0e). In this table the flag bites harder than in Table I, because the classification is made on $\tau_1$, which is exactly the column cotenancy inflates. Three of the five $\tau$-high rows carry it, and so does the one step-cost attribution the procedure makes—AIMD at its published parameters. That row is flagged by the window but not measurably damaged: its $\tau_2$, which does not touch the sampler, is 24.4 ms, inside the clean family's range. We report it as flagged rather than calling it clean, because the window is what the audit establishes and the absence of damage is an inference from a different column. The two $\tau$-high rows that are not flagged (`running gate`=4 is a watch, all-throttles-off is clean) are the ones whose elevation no window explains.

4 of the 12 rows are attributed: three to unused memory (`single-flight off`, the charge-free base, and no-gate-no-reservation), one to step cost, five have a $\tau$ no matched comparison can confirm, and three are residuals—their occupancy and their step cost both sit close to their comparison cap's, and their shortfall is real but we cannot say from these measurements which mechanism produced it. Two limits on what that buys, both of which we want read before the number is quoted. First, the comparison is a static cap matched on *wall clock*, and the arms that finish near 260 minutes are matched against `cap5`, whose SLO is 27.1%—so the classification describes an arm relative to a wall-clock peer, not relative to the static *front*, and it cannot say why the arm fails to beat the front. Second, the three unused-memory rows run fewer requests than their comparison, which is a description of what they did, not a demonstration that doing otherwise would have won; §V is the section that establishes the loss, and this table only sorts it. We publish the table rather than a summary because a summary would claim more than we have—and because "two mechanisms explain the negative result" would be the wrong summary. We also flag what changed here, since an earlier version of this table claimed four attributions from eleven rows and that was an artifact of two errors at once: the comparability clause above was not enforced in the code, and six arms had no $\tau_1$ because the run list behind this table was incomplete. The generator is now the only source for these

rows (`attribution.csv`), it enforces the clause, and it carries the co-tenancy flag rather than having it written in by hand—which is how the earlier version came to mark two clean runs.

### D. Route 1: the accounting base

Our controller's projection (Eq. 1) charges a session for its full context even while it is parked in a tool wait, when its KV is evictable and therefore absent from the engine's reported occupancy. We argued this was a double charge and removed it, admitting on engine telemetry alone.

The charge is not double, and removing it does what the argument predicted on the engine side: median occupancy rises from 49.2% to 56.2% of the pool and the busy-hour running count from 1.35 to 1.48, and the round-1+ prefix hit rate falls only from 80.6% to 79.7%, so no eviction spiral occurred. The run also finishes 25.4 minutes *earlier* (265.9 against 291.3).

It is nonetheless not a win. The SLO falls from 89.8% to 85.9%—3.90 pp— and the static front itself falls 3.81 pp over those same 25.4 minutes, so the arm keeps its place relative to the envelope to within 0.09 pp while moving 25 minutes along it: it slides, at the front's own slope, rather than crossing. What makes that a test rather than a coincidence is what it would have taken to cross: at 265.9 minutes the front sits at 87.2%, so the arm would have had to give up no more than 2.6 pp of SLO instead of 3.9, and it did not.

We are careful not to rest anything on the 0.09 pp. It is not a resolution claim, and the base configuration's own four runs show why: against this same front they sit at $-1.24$, $-1.73$, $-1.49$ and $+0.18$ pp, a 1.9 pp spread. That is the scale this statistic moves on, and it means the charge-free arm's distance from any *one* of those runs is not resolvable—measured against the second of them it is 0.40 pp *better*, against the fourth 1.5 pp worse. The slope comparison is the part that survives, and it is the part the conclusion rests on.

We want to be explicit that this is a revision. Our earlier measurement of this route had the arm 4.2 minutes *slower* and 6.0 pp worse, and we read the extra concurrency and the extra step cost—$\overline{R}$ up 1.35 to 1.48, $\tau_2$ up 23.8 to 31.3 ms—as cancelling. That measurement was one of the co-tenancy-flagged runs (§IX-0e), and its $\tau_2$ was the inflated term: measured alone the same arm reads 24.8 ms, so the second factor is $+4.2\%$ rather than $+31.5\%$, and busy-time throughput ($\overline{R}/\tau$) rises 5.1% where the flagged measurement had it falling 17%. Nothing cancels. The occupancy half of the argument (49.2% to 56.2%) is a volume measurement and was never in doubt. We keep the route in the negative column, now on the stronger ground that the highest-leverage knob the mechanism argues for moves the arm along the front rather than over it, and we record the correction here rather than in a footnote because it is the one place in this paper where a re-run changed a conclusion's reasoning rather than its precision.

We note that this claim had already been falsified once by source inspection: SGLang's `token_usage` subtracts evictable blocks, so a parked session's prefix is genuinely not counted, and charging for it reserves capacity that really will be needed back. The end-to-end measurement agrees with the source reading.

### E. Route 2: the growth reserve

Sessions grow by a median of 2.7K new tokens per round (`input_len` over rounds $\geq 1$ in the trace), so the controller reserves capacity for the next `horizon` rounds. We swept the horizon over $\{0, 1, 2, 3\}$ and the reserve is not a free parameter to be minimised: Fig. 6 shows the response is non-monotone. Removing the reserve entirely costs 12.6 pp of SLO, and a reserve of two rounds is 15 minutes *slower* than three. The optimum is at one round, and there it comes closest to the static front—0.93 pp below it, inside the tie band—and that single point is the one this paper spends the most words on, because it does not reproduce: the repeat of this configuration lands 5.6 pp below the line (§V). Read the sweep as a shape, not as a result at its minimum.

### F. Route 3: the hard ceiling

Lowering the budget fraction is the crudest available knob, and on the two settings we swept it does not behave like the others: budgets of 0.85 and 0.90 finish in 333.9 and 315.9 minutes at 89.9% and 87.7% SLO, which is $-2.24$ and $-4.22$ pp from the envelope at those wall clocks (`budget_sweep.csv`), against $-0.93$ and $-1.24$ for the two nearest arms at the 0.95 the rest of the paper uses. So this knob slides the point *below* the front rather than along it. We report it only for completeness, since it adds no mechanism.

### G. An independent adaptive controller

Every controller above is ours, which invites the objection that the negative result is a statement about one design rather than about adaptive admission. We therefore implemented a second, mechanistically different controller: an AIMD window over an agent-level admission set, following the control law of the congestion-based system we compare against in §VIII. The window grows additively while cache occupancy sits below a low watermark, is cut multiplicatively when occupancy exceeds a high watermark *and* the prefix hit rate falls below a threshold, and holds otherwise. Cuts pause the youngest sessions at their next step boundary; growth resumes paused sessions first. Every input is a runtime gauge—occupancy and hit rate from the engine, nothing from the trace—and the window acts through the same named permits as Foyer.

It does not beat the static front either, at either parameter setting (Table I). At the values reported in that paper's evaluation it finishes in 364.3 minutes at 81.4% SLO, 11.2 pp below the envelope; the same arm with its retuned watermarks finishes in 264.6 minutes at 52.8%, 34.3 pp below (`table1.csv`). The retuned setting is also one of the most volatile runs we report: its segment-to-segment SD is 13.2 pp of SLO, with segment values swinging between 37 and 72. Two runs are more volatile (`cap4` at 15.8 and the all-throttles-off arm at 14.4), and no other arm that keeps its throttles on is as bad—which is what a control loop oscillating around a capacity cliff looks like.

We report two things honestly about this comparison. First, this is our reimplementation of a published control law, not the authors' system, and a reimplementation of any controller can underperform its paper. We built it to the published law, gave it the same permit interface and the same 2 s cycle as Foyer, and did not tune it beyond the two documented settings; but the caveat stands. Second, both controllers stall. At its paper parameters this one pinned its window at one session, admitted nothing, and left the engine idle for over an hour. Our own controller does the same thing more briefly: in 4 of the 11 Foyer runs a scan finds windows in which the engine was idle, 85 or more sessions were waiting, the SLO valve was open and the controller issued no permit at all. Three of those runs have a single such window each—170.5, 173.2 and 191.5 s, 0.9–1.1% of the run's wall clock—and the arm that disables every throttle has nine of them, together 12% of its wall clock (`stall.csv` states the criterion; an earlier draft quoted 150–175 s and 0.9% for all four, which no version of the criterion reproduces). The remaining runs, including the best one, show no such window. Nothing is logged as an error and the run recovers on its own.

We traced one of those windows to a code path that explains it: single-flight admission considered only the head of the FIFO queue, and when the head was a session too large to fit the remaining budget it was never admitted, never left the queue, and blocked every smaller session behind it. That is the code every run in this paper executed. The fix in our repository—scan the queue, still admit at most one per cycle—is *not* in the artifact that produced these results, so the numbers here predate it and we do not claim them for it. We are also explicit that this is *a* path, not a proven cause: the audit is a log signature, and the same signature can be produced by other closed gates. What we can say is that the signature is present in a third of our Foyer runs and absent from all of our static caps, which have no admission decision to get stuck in.

That is the honest version of a finding we first reported too strongly. An intermediate draft of this paper called it a deadlock and said the configuration that produced our best result was the one that deadlocked on its repeat. Neither holds: the windows are minutes, not hours, and they clear on their own, and the configuration that produced our best result never showed one. Nor do they move any verdict we report—0.9–1.1% of the wall clock in each of the three runs that has a single one, and in the all-throttles-off arm 12% of a wall clock that is already the slowest in the table. We keep them because the mechanism is real and its size is the point: on a pool that holds four median sessions, a controller's failure mode is a silent stall, not graceful degradation, and it took a repeat run to see it.

### *H. Route 4: execution granularity*

The first three routes all assume that what a controller can set is *how many sessions are resident*. A static cap sets the same quantity, so it is unsurprising that they land in the same place. The fourth route attacks the assumption instead: because a large fraction of live sessions are parked at any instant, a cap on *resident* sessions leaves the engine idle whenever all residents happen to be parked. Capping the *running-request* count instead should keep the engine busy while leaving the resident count high.

The permit design supports this directly: a paused session finishes its current round and blocks before its next one, and resumes when the controller re-admits it. We implemented it as a gate on the engine's own `num_running_reqs` and swept the gate over $\{2, 4\}$. Only the gate-of-four arm ran to 998 of the trace's 999 steps; the gate-of-two arm reached 770 and is not used, so what we report below is a single comparison against the no-gate arm rather than a sweep.

It does not work, and the measurement says why. Against the no-gate arm, a gate of four moved mean running requests from 1.33 to 1.30 and mean live sessions from 3.25 to 3.03: it tightened both, by 2% and 7%, rather than pinning the one it aims at. That is what we observe, and we state it as the observation it is rather than deriving a bound we cannot support. A live session spends most of its time parked in a tool wait (the ratio of mean running requests to mean live sessions is 0.33 to 0.64 across the twenty-three arms, `occupancy.csv`'s `running_mean` over `runs.csv`'s `conc`), so pausing live sessions removes capacity that was not running in the first place—but that ratio is an average over a run, not a bound on any instant, and the running count can in principle reach the live count. Both repairs fail: without a constraint on admission, a paused session is re-admitted on the next cycle and blocks for at most one 2 s cycle; with one, the policy is a session cap again. The obstacle is granularity—`paused` takes effect at round boundaries, while pinning an instantaneous count requires control the permit mechanism does not have.

The measured outcome matches: the gate moved running concurrency from 1.33 to 1.30 and live sessions from 3.25 to 3.03. It tightened rather than pinned, and a tightening is another cap.

## VI. The Same Measurements on a Second Workload

Everything so far is one trace family, and a negative result drawn from one family invites the objection that the family is the finding. We ran the same accounting, a static sweep from a cap of two to a cap of eight, and the same controller on a second replay: 117 real Claude agent sessions, 390 rounds, the same model, the same engine and the same 101,432-token pool. One difference has to be stated up front, because it bears on the paper's central variable: *this trace was run entirely without a host tier*. Its static points, its envelope and its controller are all device-only. That makes the comparison internally fair—both sides of the front are the same configuration—but it means this workload tests the negative result and not the tier mechanism, and it leaves an alternative account of the controller's failure standing: it failed here on the configuration where §III says added concurrency is expensive. The two workloads differ in ways that matter for this paper, which is why we chose it. Its sessions are smaller (median prompt 18,809 tokens against 23,187) and its tool waits are

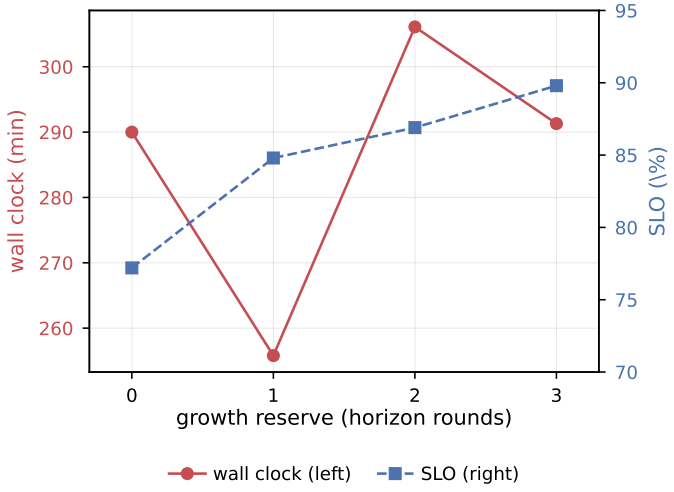



Fig. 6. The growth reserve is not a free parameter. Removing it costs 12.6 pp of SLO; two rounds is 15 minutes slower than three. The optimum at one round comes closest to the static front, 0.9 pp below it—and is the one point here that does not reproduce on a repeat (it lands 5.6 pp below). The other three settings were each run once, so the shape is better supported than any single point on it.

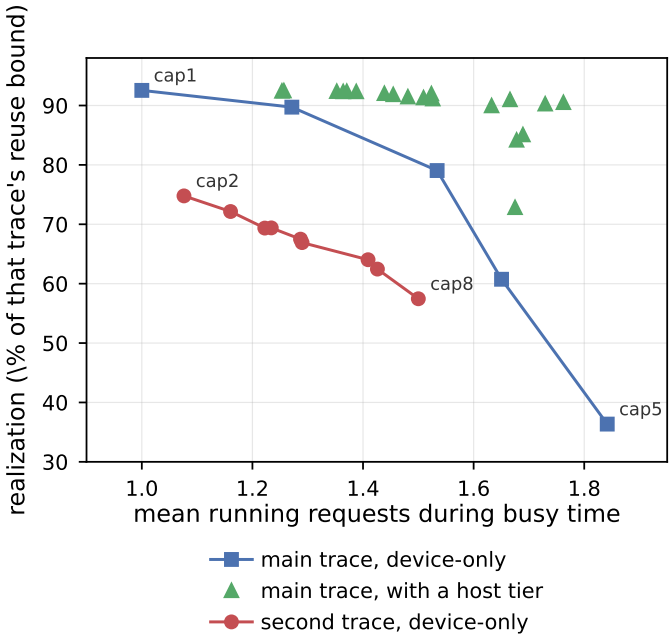



Fig. 7. Realization against the running count on both traces, each normalized to its own trace's reuse bound. Device-only, the main trace falls from 92.5% to 36.4% and the second from 74.8% to 57.5%; with a host tier the main trace's runs stay at 72.8–92.5%. The second trace is lower throughout—interpolating the main trace's line at the second trace's running counts gives a 17-point gap at the lowest and 23 at the highest—and it declines more gently, which is what "the direction transfers and the level does not" looks like on this axis.

far longer at the median: over round transitions—same session, adjacent rounds, each session's last round excluded because it has no successor—the gap is 20.2 s against 0.34 s at the median—about 60×—and 39.9 s against 15.8 s at the mean, which is 2.5× (`toolwait.csv` carries both traces on this definition). The 29.1 s that appears in `elasticity_x.csv` is the mean over all 390 rows including each session's last round, and is not comparable with the main trace's 15.8 s; we use the transition definition throughout §VI. So the engine is idle far more of the time, and the running counts are lower.

### A. What transfers, and what does not

*a) The reuse accounting transfers in direction, not in level.:* The second trace plans a higher reuse bound than the first—86.3% of its prompt tokens against 75.5%—and realizes less of it, at a lower device occupancy (Fig. 7). Realization falls from 74.8% to 57.5% as the running count rises from 1.08 to 1.50, so the direction of §III survives. What does *not* survive is the sharpness. On the main trace the decline is a threshold: the per-unit slope is $-10.4$, $-40.6$, then $-157.1$ and $-127.6$ points per unit as the running count passes 1.53, which is the cliff §IV measures. Here the slope stays gentler through the same region—$-31.2$, $-36.8$, $-28.4$, $-93.0$ and $-67.6$—and the two steep numbers at the end sit on spans of 0.017 and 0.074 of running count, where a difference between two runs starts to read as a trend. Taken as a whole the sweep loses 74.8–57.5% of this trace's bound between $\overline{R}$ 1.08 and 1.50 at $-40.9$ points per unit. We do compare the two traces' overall slopes, which is the one slope comparison the data supports because both are taken over their full measured ranges (`slope.csv`): the device-only family runs from 92.5% to 36.4% over $\overline{R}$ 1.000–1.842, a whole-range slope of $-66.8$ points per unit, and this trace from 74.8% to 57.5% over 1.08–1.50, a slope of $-40.9$. So per unit of running count, the main trace loses its bound about one and a half times as fast over its range ($-66.8$ against $-40.9$)—but the comparison is between ranges of different length and different absolute level, and we do not lean on it. What this trace does establish is that the *level* does not transfer, and we treat that as a bound on what we can claim: it realizes *less* at 15.8% median device occupancy than the main trace does at 22.4%—the two numbers are the lightest cap on each trace (`realhr_cap2` realizing 74.8% and `cap1` realizing 92.5%), both medians of `sglang:token_usage` with means of 13.9% and 25.2% (`reuse_x.csv`, `occupancy.csv`)—so realization is not a function of occupancy alone and a rule of the form "the pool collapses above $x$% occupancy" is false on our own data. Occupancy is instantaneous; reuse depends on whether a prefix survives from one round to the next, which is a history the instantaneous number does not carry. We do not measure that history here, and we say so rather than fitting the two traces to a common curve.

*b) The elasticity criterion transfers in sign, not in place.:* On both traces the criterion plays the same role, and it is the modest one §IV settles on: it *rules out* the workload-side story. The controller on this trace runs at a busy-time mean of 1.223, which falls inside the span its own static caps cover (1.08 to 1.50)—more precisely inside the cap3→cap4 segment, where $\varepsilon(\tau_1) = -0.28$ over $\overline{R}$ 1.16–1.29, so its failure is again not explained by the criterion. What the completed sweep adds is

that this trace does approach the criterion's *expensive* branch, later and more gently than the main trace: above $\overline{R} \approx 1.29$ the model-consistent elasticity turns positive and touches one in the first of the three segments there ($+1.0$, then $+0.18$ and $+0.87$ as the sweep reaches 1.50), while the main trace crosses one at 1.53 with $+1.15$ and stays above it. So the direction is not a property of one family; the location is. We report the point estimates and not a threshold, because a $\pm 10\%$ perturbation of $\tau_1$—inside the range our repeat pairs show—carries each of those three segments' bounds across zero *and* across one. The per-request measure is no steadier ($+0.45$, $-1.25$, $+0.73$). And the controller never operates where the branch turns expensive: 1.22, below 1.29 and below the main trace's 1.53 alike.

We also record what the noise does, because it is the same weakness the whole paper has. Recomputing each segment's $\varepsilon$ with $\tau_1$ perturbed by $\pm 10\%$—against the 0.3–1.1% our clean repeat pairs differ by in $\tau_1$ (`decomp.csv`; one anomalous pair reaches 42.5%), so the perturbation is generous rather than conservative—gives bands of $[-5.20, +0.41]$ for cap2→cap3, $[-2.17, +1.61]$ for cap3→cap4, $[-1.25, +3.26]$ for cap4→cap5, $[-14.07, +14.43]$ for cap5→cap6 and $[-3.33, +5.07]$ for cap6→cap8. Every one of them crosses zero, and the three at the end cross one as well. So at that perturbation no segment's sign is resolvable, and the turn the paragraph above describes is a property of the point estimates rather than a boundary we have located—which is a bound on that paragraph, not a reprieve from it.

*c) The controller's verdict transfers, weakly.:* Foyer finishes the second trace in 75.6 minutes, against 66.2 for the fastest static run on it (the cap of eight), so Foyer is 14.3% slower (14.26% at full precision). We do not report that as a loss. The repeat pairs we have vary by 1.8–4.7% in wall clock (`replicates.csv`), and this trace gives us two of its own on the same clock (`repeat_spread.csv`): the cap of four repeats to within 1.6%, and Foyer itself to within 0.02%, the latter accompanied by 0.03 points of realization on the volume axis. The gap is more than eight times the wider of those, so it is not repeat noise on either side of it—but two pairs cannot bound the variability of the *other* configurations in the comparison, and run-to-run spread on this machine reaches 22.0% across the other configurations and tiers we have measured (`repeat_spread.csv`; 0.02–6.08% across the ten repeat *groups* those two traces give us—four runs in one of them, two in each of the rest—the spread between the groups themselves being the reason we report a range and not a number). So the honest reading is the one we have used throughout: at the threshold this paper uses, it did not win on either axis. Its signed distance from the second trace's static envelope is $+0.40$ pp at 10 s on the SLO axis—inside the $\pm 1.0$ pp tie band, which is defined on that axis—and the repeat of the same configuration reads $+0.40$ as well ($+0.4012$ and $+0.4017$ at full precision), so this is not a single measurement standing alone—and $-0.14$ pp on the cache-hit axis, against the 14.9 points that separate this trace's own six static caps on that axis. That axis is threshold-dependent here, and in the opposite direction from the main trace's: at 5 s this trace's caps separate much more (the cap of eight scores 78.2% against the cap of two's 99.0%) and the controller sits $+1.87$ pp above the envelope—outside the band, nominally a win on that axis, and the paragraph's last sentence takes it back—whereas on the main trace none of the cleanly measured controllers reaches the front at 5 s, and one appears above it at 20 s in a single measurement whose repeat lands below the line (§IX). We report both readings rather than the one that gives a single answer, and we report one more thing the repeat settled: *that single crossing does not reproduce*. The same configuration run again reads $+0.86$ pp at 5 s—inside the band—against the first run's $+1.87$, while at 10 s the two agree to three decimals ($+0.40$ each). So this trace's one crossing is also a place where a second run puts the controller back inside the tie band; §IX reports the same outcome for the main trace's, on its own repeat. (The envelope and the elasticity below are both built from all six static caps the sweep now has; the three caps the sweep added after the first version of this comparison—the caps of five, six and eight—are the reason the last three segments exist at all.) The SLO number is nearly uninformative, and we flag that rather than let it stand as a measurement. At the three caps the envelope was originally built from, every point on this trace scored 99.5–100.0%, so the axis could not discriminate at all. The three the sweep has since added do separate—the cap of eight scores 93.8% against the cap of two's 100.0%—so the axis is not flat after all, but its whole spread over six static caps is 6.2 points against a $\pm 1.0$ pp tie band, and Foyer sits at 100.0% on it.

### B. What this section does and does not buy

It buys a negative result that is not a property of one trace family: on a second workload, with a different session-size distribution and a tool-wait distribution about $60\times$ longer at the median, the same controller still does not beat the same kind of static cap, and the reuse degradation still appears. It costs us the transportability of the threshold. On the main trace the criterion narrowed the search by excluding the workload-side explanation and put the turn of the point estimates at $\overline{R} \approx 1.53$; on this trace the same measurement turns positive above 1.29. What we cannot do is *locate* either turn—the $\pm 10\%$ perturbation puts every segment's bounds across zero on both traces—which is the caveat §IV already states for the main trace, now with a second trace behind it. The controller is below both turns (1.22), so the criterion does not explain its failure on either trace. We report both, and we note that the second workload is closer to the regime where adaptive control *should* pay—lower concurrency, much longer idle gaps—and the controller still did not win there.

## VII. Load Dependence

A negative result about adaptivity invites the obvious objection: adaptivity pays when conditions *change*, and a single stationary trace is the one case where a static cap should be near-optimal. We agree. What we test here is the weaker thing—two separate constant-load tiers—and we also ran the

stronger one, a trace whose load moves mid-run, whose baselines co-tenancy made unusable; §IX reports both what this section leaves open and why that experiment has no result to give.

We run the same controller and the same cap family on a 25-session tier. The throttle configuration is reversed: with all safety throttles disabled, the 25-session tier runs *51% faster* (58.4 to 28.8 minutes) at a cost of only 2.4 pp of SLO, while on the 200-session tier disabling the same throttles costs 42% (291.3 to 415.0 minutes) and 43.7 pp of SLO. One switch, two regimes, opposite signs, both around 40–50% (Fig. 8). No single scalar setting is optimal across both.

We flag the generation gap rather than let the figure imply more than we ran. The two 200-session numbers are current; the two 25-session numbers are from an earlier harness revision, and we never ran the all-throttles-off arm on the 25-session tier under the current one. So the reversal is a direction we measured across two harness generations, not a same-generation paired comparison, and a reviewer should read it that way. The same-generation 25-session measurement we do have is in Table II, and it does not contain that arm.

This is the regime in which adaptive control should win, and we now have the same-generation measurement that earlier drafts of this paper were missing. Four arms on the 25-session tier, all under the current harness (Table II): static caps of two, three and four finish in 32.4, 24.2 and 23.6 minutes at 92.8%, 91.2% and 79.2% SLO, and Foyer finishes in 27.6 minutes at 92.8%.

The finding is a trade, not a defeat, and it is the opposite of what the 200-session tier shows. Against the same-generation envelope—which runs from `cap4` at (23.6, 79.2) through `cap3` at (24.2, 91.2) to `cap2` at (32.4, 92.8)—Foyer sits $+0.9$ pp above the line. A static cap of three is still 12% faster in wall clock, but it gives up 1.6 pp of SLO to do it, so neither dominates the other. On this tier the controller is a different point on the front rather than a dominated one, which is exactly what the 200-session tier denies it.

We are careful about the size, and about the yardstick. $+0.9$ pp is measured against a straight-line interpolation between static points, not against a switching policy anyone ran, so the defensible statement is that Foyer is not dominated by any static point we measured on this tier—not that it beats what a static mix could achieve. On 125 requests one request is worth 0.8 pp, so the margin is about one request. The tier's segment-to-segment SD (5.1 to 21.4 pp) is not the right yardstick: segment volatility is not run-to-run uncertainty. The right one is the replicate spread we have—which, across the five configurations we repeated, is 0.5, 0.8, 1.0, 3.9 and 5.1 pp (`replicates.csv`). Against the two tightest of those $+0.9$ pp is about a replicate wide; against the three widest—which belong to the configurations that decide our headline elsewhere—it is inside one. We quote the tightest pair because it is the least favourable to us, not because it is the whole distribution. What we will defend is the weaker claim: on the low-load tier the controller is not worse, and the same switch that is worth $+51\%$ there costs $-42\%$ at high load. A

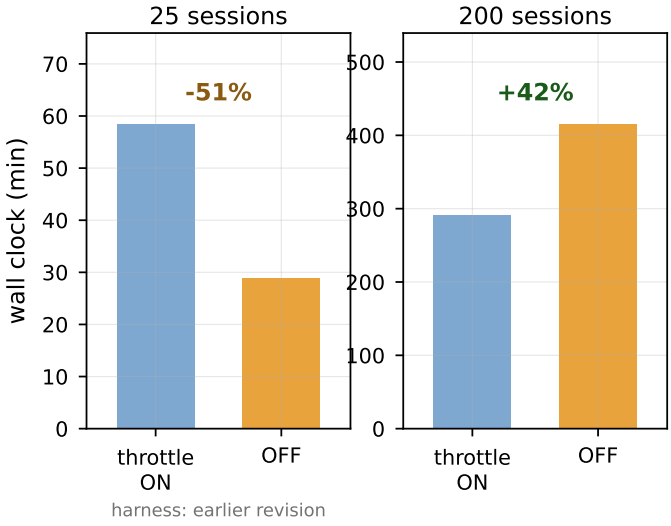



Fig. 8. The same throttle switch, two load tiers, opposite signs. The 25-session bars are from an earlier harness revision and are shown as a direction, not a magnitude—the current-harness 25-session comparison is in Table II instead. The 200-session bars are current. The two panels have independent axes, so the bars are not comparable in height across panels—the annotations give each panel's own change.

controller that is competitive where load is light and dominated where load is heavy is the pattern this paper set out to explain, and it is the pattern we measured.

The second finding is the more useful one. The signal an adaptive controller would need is available—the number of sessions waiting for admission is four times larger on the 200-session tier than on the same-generation 25-session arm (median 66 against 16, maximum 135 against 25; the earlier-harness 25-session arms of §VII read 9–11, which is where the 6–7× of an earlier draft came from, and we quote the same-generation pair)—but its *interpretation* is not obvious. At the 25-session tier the controller holds sixteen sessions back while running one; that backlog is manufactured by the throttling itself rather than evidence that demand has slackened. Reading it correctly requires combining instantaneous backlog with progress through the run, which is a stateful computation a cap cannot perform. We regard this as the most promising direction we did not finish, and we are explicit that we did not finish it.

## VIII. Related Work

*a) Admission and scheduling for agentic serving.:* Recent systems report large gains from treating an agent session, rather than a request, as the schedulable unit, and from preserving session state across tool calls: session-aware admission with CPU-restorable KV [5], [6], program-level pause/resume with shortest-context-first restoration [7], and congestion-based control over an agent-level window [8]. A recurring measurement in this line is that recomputation, not admission, dominates: [8] attributes 49.1% of end-to-end latency to recomputation, and [9] preserves state on the CPU at the cost of a reload. Our results are consistent with this

literature rather than in tension with it, and §IX explains why: the mechanism these systems add is one our controller already has.

*b) Engine-side techniques.:* A separate line improves the engine rather than the admission decision. Iteration-level batching and paged KV [4], [10] are the substrate every policy here runs on; chunked prefill with stall-free batching [11], prefill/decode disaggregation [12], and cache-centric scheduling [13] are the refinements. Our §III measures 2.6–8.2 ms of each 27–37 ms step as work that is not decoding—prefill and non-decode overhead—which is exactly what this line attacks. We expect these techniques to move the frontier rather than to decide who sits above it, but we did not run the interaction and so state it as an expectation.

*c) State retention and preemption.:* A third line keeps session state alive across the gaps we measure in §III: keeping KV on cheaper storage across turns [14], [15], migrating it between instances [16], and preempting at iteration granularity [17]. These systems exercise finer control than our 2 s admission cycle, which is the boundary of our negative result: we falsify one control granularity, not the family. What they share with Foyer is that the state is not discarded—which is why, as we argue below, adding admission intelligence on top buys nothing here.

*d) Application-level dependencies and replay.:* Serving an LLM *application* rather than a request means the scheduler sees dependencies between calls [18], and agentic traces have structure that request-level benchmarks do not capture [19]. Our replay preserves that structure but fixes it in advance, which is exactly the limitation these works would press on.

*e) Prefix caching.:* The prefix cache is often proposed as the signal an admission policy should use. What we measure is that the *realized* reuse is a function of the admission policy and of the host tier, while the *planned* reuse is a constant of the trace; an earlier draft of this paper conflated the two and got the direction wrong in both. The trace plans a round-1-and-later hit rate of 80.9%. Without HiCache the realized round-1+ rate follows the plan only at the loosest cap and then collapses: 80.8%, 78.5%, 70.6%, 56.0%, 32.8% for caps of one through five. With HiCache it tracks the plan at every cap we ran: 80.7%, 79.6%, 78.7% for caps of two, three and four. Our controller sits at 80.6%. So the honest statement is not that admission does not move this quantity—`cap4` alone moves it 25 points—but that the host tier is what keeps it from moving, which is the same mechanism §III and §IV keep running into. A configuration without a host tier that admits four sessions does not have the cache that a host-tier one at the same concurrency does. All eight numbers in this paragraph are the *observed* per-step hit rate (`server_prefix_hit_rate` in the per-step logs), unweighted mean over rounds $\geq 1$; the planned rate is the trace constant 80.9% and does not move with the cap. §III uses a different convention—all steps, token-weighted, so that reuse and recomputation partition the prompt—under which the same collapse reads 92.5%→36.4% of the trace's bound for the device-only caps and 92.4%→90.2% for the static host-tier caps. We state both because the paper's earlier drafts mixed them. We note the boundary of this claim—we varied admission policy, not eviction policy, and SGLang exposes alternate eviction policies we did not test. A characterization of production KV-cache traffic at a large cloud provider finds that eviction-policy design is "highly workload-dependent" and that reuse patterns differ sharply by request category [20]; that is reason to expect a single trace family to understate the space, though it is not a policy comparison and we do not present it as one.

## IX. Discussion

*a) Why published gains do not reappear.:* The systems cited above win by keeping session state alive across tool waits and by pausing at agent granularity rather than request granularity. Foyer has both properties by construction: the permit file gives per-session pause and resume, and the controller never frees capacity by discarding a session—it admits fewer sessions instead. To that extent the mechanism these systems add is one our controller already has, which is one reason adding admission intelligence on top of it buys nothing here.

We are careful not to overstate this, because the distinction is between *choosing* to discard state and *having* it evicted. Retention here is the engine's radix cache doing its ordinary job; it is not something the controller implements, and §III measures that the cache evicts parked prefixes under pressure and that they are recomputed. What Foyer avoids is the deliberate trade of state for capacity. A system that manages the host tier as an explicit policy—as [9] and [14] do—holds a lever we do not, and our negative result does not cover it.

*b) The threshold decides the verdict, and we report that.:* Every SLO number in this paper counts a request as good if its time-to-first-token is under 10 s. That constant sets the whole vertical axis, so we re-derived the frontier at 5 s and 20 s as well, from the same per-request logs. The ordering of the static caps does not change. What changes is where the controllers sit relative to them, and at the loose end one of them appears, in one of its two runs, to cross:

| SLO threshold | 5 s | 10 s | 20 s |
|---|---|---|---|
| `horizon`=1 | −2.9 | −0.9 | +1.5 |
| `horizon`=1 (repeat) | −6.8 | −5.6 | −2.3 |
| Foyer base | −2.3 | −1.2 | +0.6 |
| `bres` | −3.6 | −1.3 | +0.3 |
| Concur-style AIMD | −14.2 | −11.2 | −6.5 |

Two readings, and we give both. The first is that our headline claim is conditional: at the strict threshold that interactive serving actually cares about, none of the cleanly measured controllers reaches the static front; at a loose one, one measurement of one of them does. The second is that the crossing is not a win, and there are now two independent reasons to say so. The first we gave an earlier draft of this paragraph: the margin is not resolvable. +1.5 pp at 20 s is larger than the replicate SLO spread of the three tightest pairs we have—0.5 pp for the base controller, 0.8 pp for `cap3`+HiCache and 1.0 pp for `cap2`+HiCache—but smaller

than the two widest, 3.9 pp for the best arm's own repeat and 5.1 pp for `cap4`+HiCache, and those two are anchors. It is the anchors, not the controllers, that set the line the margin is measured against, so the comparison that matters is the one against the widest anchor—where the margin is not resolvable. The second reason is harder to argue with, and we found it only when the arm's repeat landed: *the crossing does not reproduce*. `horizon`=1 reads $+1.5$ pp at 20 s and its own repeat reads $-2.3$—not merely inside the band, but below it, on the other side of the line the first run crossed. Two runs of one configuration, 3.8 pp apart on this axis ($+1.5$ against $-2.3$, from 94.4% and 91.1% of requests served under 20 s), and the sign of the verdict flips with them. So the loose-threshold crossing is a single measurement, and the one arm whose repeat we have on the axis where it matters does not support it. (An earlier revision discarded the `cap3`+HiCache pair on the ground that one run had exceeded its cap. That was wrong: the over-cap run is the *device-only* `cap3` repeat, and both HiCache runs peak at three with none over. The pair is a valid replicate and we use it, including as an envelope anchor.) A margin that appears only at one threshold and only at a size we cannot resolve is not evidence that adaptive admission works; it is evidence that the question needs more runs than we have. We state the conditional version in the abstract, and we flag this as the second thing we would re-run with more GPU time.

*c) The SLO metric excludes the controller's own cost, and we checked.:* Every SLO number above counts a request as good if its time-to-first-token is under 10 s, and the runner measures that from the moment the request is *submitted to the engine*. A session that waits for a permit has not been submitted yet, so the wait is invisible—and that wait is precisely the cost an admission controller imposes. We therefore recomputed the runs we have per-step logs for on a ready-to-first-token basis, where a round's clock starts when the session is actually able to run (its arrival for round 0, the previous round's completion plus its tool wait otherwise). That is nineteen of the twenty-five 200-session runs in Table I; absent are the no-control baseline, the two HiCache anchor repeats and three controller repeats, and we do not extend the claim to them.

The correction is large. Every configuration loses 2.9–19.3 pp of SLO, and the loss is systematic: tight caps queue the hardest, so `cap1` falls from 94.8% to 76.2% while `cap5` falls only from 27.1% to 24.2%. A session admitted last under `cap1` can wait hours for its turn; its engine-side TTFT is still 1.6 s.

What does *not* change is the verdict, within the subset we recomputed. The static envelope keeps the same four anchors—the two anchor repeats, which we did not recompute on the ready clock, stand in as their first runs, and every distance below carries that substitution (`arrival_slo.csv` records the columns)—and of the eleven controller arms, nine move by no more than 2.5 pp. Two move more—the all-throttles-off arm improves by 8.2 pp and the retuned AIMD arm by 3.2 pp—and both stay below the front either way. One crosses the $\pm 1.0$ pp band, and it is the one that decides our headline: `horizon`=1 goes from $-0.93$ (inside the band, a tie) to $-1.21$ (below it), because on the ready clock its own SLO falls from 84.8% to 68.2% while the anchor it is measured against falls less. The ready clock therefore moves it from a tie to a loss, which is the same direction the paragraph above reaches from the other side, and it is why we do not report a ready-clock distance for it as a headline: the two anchor repeats that define the line are not in the set we could recompute (§IX says which runs are), and inventing a line for them would be the same error this subsection is about.

The subset matters and we state its edge. The two configurations whose repeats decide how we read the verdict—`horizon`=1's repeat and the fastest anchor `cap4`+HC's—are not in the nineteen, because we did not recompute their per-step logs on the ready clock; the rebuilt envelope therefore rests on the original anchors without their repeats, and a ready-clock frontier that included them could move the line by the 3.9 and 5.1 pp those repeats differ by. We do not claim the verdict is unchanged under every SLO definition—only that on the runs we could recompute, moving the clock to the ready point costs every configuration 3–19 pp, moves no configuration from below the line to above it, and changes exactly one controller verdict: `horizon`=1 leaves the tie band on the below side. (One static cap crosses the other way, `cap3`+HC from $-1.72$ to $-0.79$, which does not help any controller.) We report this because a reviewer should ask it, and because the answer is the useful part: the negative result is not an artifact of choosing an SLO definition that flatters controllers. If anything the flattering choice was the one that moved the arm nearest the line from $-0.9$ to $+1.5$—that is, from just below the line to just above the tie band's upper edge, by a margin of the same order as that pair's own replicate spread (3.9 pp).

*d) Limitations.:* Our results come from one 24 GB accelerator and one 7B model. The workload axis we did test once more (§VI); the hardware axis we did not, so generality across devices is untested. Only five configurations have been repeated—four of them twice and the base configuration four times—so the replicate spread we report in §V is estimated from one to three degrees of freedom per configuration; a margin as small as the one that decides our verdict would need several more repeats per configuration to resolve, and we did not run them. Two specific boundaries matter. First, the pool holds about four median sessions, so concurrency is capacity-bound by construction; on a device with a much larger pool the premise may not hold and the conclusion may invert—we regard reproducing on a larger device as the single most important next experiment. Second, the load is stationary, single-tenant and single-SLO. The load-dependence result of §VII is measured on two *separate* constant-load tiers, so it says what to expect when conditions differ, not what happens while they change. We did run the experiment that would settle that—a trace whose load moves mid-run, with a static cap on each tier and a switcher that changes its cap at the trace's designed switch point—and we cannot report a result

from it, in either direction, because its static baselines were measured under host co-tenancy: `var_cap3` spent 158 of its 266 minutes at four or more concurrent engine instances, and `var_cap2` is flagged as a watch (§IX-0e). A first look at those runs has the switcher finishing 15 minutes ahead of the best static cap at an SLO inside the tie band, which would be the first evidence in this paper that adaptivity pays; we do not report it as a result, because the baseline it beats is precisely the one the audit flags, and a 60%-contaminated baseline is inflated in exactly the direction that manufactures that finding. Clean re-runs of that family are in flight, and the honest position until they land is that the variable-load case is measured and unusable, not unmeasured. Multi-tenant and heterogeneous-SLO settings are where we would look next.

*e) Reproducibility.:* The front and the signed distance of each controller are computed by `scripts/lab/frontier.py` from the raw per-step logs; Table I is rendered by `scripts/lab/gen_table.py` from `frontier.py`'s full-precision coordinates (`paper/data/runs_precise.csv`), including the four runs that stopped short of 999 steps; the live-session column comes from the runner's admission log via `run_summary.py`, and §IV is computed by `gen_elasticity.py` and `gen_elasticity_hc.py`. Every run is guarded against a shrunken KV pool.

*Co-tenancy on the measurement host.* Our runs share a host with other tenants and with each other, and late in the campaign we found that this matters more than the pool guard catches. An audit (`gen_concurrency.py`, data in `paper/data/concurrency.csv`) reconstructs, for every run, how many other runs were alive for how long. Runs that overlapped three or more others for at least an hour are flagged; fourteen of ours are, two of which are failed runs we do not report. The flag is a risk marker, not a measurement of damage, and we treat it that way: five of the twelve (`F:t85`, `F:t90`, `var_foyer`, `cap1` and the published-parameter AIMD row) show no aggregate inflation at all, because their overlap was short relative to the run or did not touch their decode interval. Where the damage is legible it is large—the per-request decode interval $\tau_2$, measured client-side and never touching the sampler, sits at 30.9–33.3 ms for the three affected arms still carrying the flag in Table I against 23.7–25.0 ms across the seven clean runs of the same controller family—a 24 to 41% inflation. A fourth affected arm has since been re-measured alone and came in at 24.8 ms, inside the clean range, which is the correction in action. Two controls pin the cause to co-tenancy rather than to the configuration, and both are within-run—the same run, the same policy, the same trace, with the only change being who else was on the machine. One run (`cap1`) reads a per-half-hour median $\tau_2$ of 20.3–20.9 ms for its first five and a half hours, and 21.3, 27.1 and 21.1 in the three half-hours after two others start. Two runs move the other way as neighbours leave: `F:t85` from 29.8 ms in the half-hour before to 22.3–22.6 in the four after, `F:rl99` from 32.4 to 25.2 and 25.1. We chose the half-hour window because it is the widest one that resolves the transition and still puts at least twenty steps in every bucket quoted; the finer series is noisier and does not support a minute-resolution claim, so we do not make one. (`tau2_windows.csv` carries the series.) Fourteen of our runs fall in the flagged class; five of them are rows of Table I (`cap1`, `F:hz0`, `F:hz2`, `F:breshz1`, and the published-parameter AIMD row) and nine are arms we do not report here. The flag is a window, and only three of those five are actually damaged—the controller variants whose $\tau_2$ is 30.9–33.3 ms. `cap1` and the AIMD row show no aggregate inflation (20.6 and 24.4 ms), so we keep their numbers and flag the window. A sixth row carried the flag and the damage, and has been re-measured alone under the same one-run-per-GPU discipline: `bres`, whose clean $\tau_2$ is 24.8 ms. Its row in Table I now carries the clean numbers, and the two conclusions that rested on it—its step-cost attribution in §V-C and the route-1 comparison of §V—are restated in that section rather than left provisional. The remaining three damaged rows are still being re-measured, and until those runs land we treat their wall-clock, SLO and step-cost columns as unusable: the signed distances in Table I for those three rows and the $\tau$-high attributions of §V-C where it names them. We flag them rather than quietly dropping the rows so that a reader can see which conclusions are still standing on which measurements.

Two things we can already say. First, the failure is silent: the engine-side counters—running count, pool occupancy, evictions, recomputation—are indistinguishable between the affected runs and the clean ones, so nothing in the ordinary telemetry would have caught this. Second, the affected runs' *volume* columns look untouched: the three damaged host-tier arms reuse 0.6791–0.6959 of the prompt against 0.5497–0.6981 across the sixteen undamaged host-tier arms, so the damaged runs sit inside the clean range rather than apart from it—suggestive, but not proof, since co-tenancy could in principle also reorder completions and change eviction order.

The one arm we have both ways settles the second point directly, and we keep the two runs side by side in `superseded.csv` because the superseded one is by construction absent from the tables the rest of this paper reads. `bres` measured under co-tenancy reuses 0.6943 of its prompt; the same policy measured alone reuses 0.6904, a 0.39-point difference, while its per-request decode interval moves 31.3 to 24.8 ms and its wall clock by 10%. The volume columns therefore carry no signature of the contamination that we can detect—one arm is one comparison, and we are not claiming more than that—which is why the audit had to run on the client-side clock. For the three arms still outstanding, the re-runs will confirm it policy by policy, and we compare reuse per round and per session between the pairs rather than only the totals.

One caveat about the controller artifact, which a reader with our code would otherwise hit. There is no single controller copy behind this paper: the runs record four different hashes of `controller_budget3.py` (`2b573829083c` on 31 runs, `f825bc768188` on 24, `d692ab11f396` on 4 and `e1dd5ae78019` on 3), so the artifact evolved during the campaign and a reader cannot reproduce every number from

one revision. None of the four matches the repository copy (`a21bd2b63774`), and the difference is not one hunk but several: the queue-scan fix described in §V-H, and a rename of the logging variable whose rebinding §V identifies as the source of a retraction. The repository has both; none of the four running copies has either. The numbers in this paper therefore describe the pre-fix path, with one partial exception we want to state precisely rather than let a reader assume the stronger thing. We did run the same configuration with `sf=0` (§V-B), and on the running copy that changes two things at once: it scans the queue *and* drops the one-per-cycle limit, because the `break` that enforces single-flight is inside the same conditional. The repository's fix keeps the limit and only adds the scan. So our run is not a test of the fix as written—it is a test of the fix plus a looser admission rate, and it moves the arm along the front rather than past it. The narrower experiment, scan-with-the-limit-kept, is the one we would do first with more GPU time. We report six claims we retracted during the work, because the retractions were the most informative events. In order of how much they cost us:

*The denominator of the step cost.* Our first §III divided by the whole-run mean running count while multiplying by busy time. The two are not the same quantity—the whole-run mean includes the idle samples—so $\tau$ came out low by exactly the busy fraction, 36% for `cap1`, and two independent errors cancelled to make our two measurement methods appear to agree. This is the one that changed a headline number.

*A mechanism refutation that was itself refuted*: we withdrew the eviction-to-recomputation account on the strength of two checks that were both identities rather than measurements (§III).

*A median-based reading of the tool wait* that inverted a central assumption. *A monotone relation between idle fraction and concurrency* that the full data does not show. *A growth-sensitivity conclusion* that rested on an arm we had already invalidated. And *the accounting double-charge* of §V, falsified twice—once by source inspection and once by the end-to-end measurement here.

## X. Conclusion

We asked whether an adaptive admission controller can beat a tuned static concurrency cap on a memory-bound agentic LLM workload. On our system we could not build one that does: nine variants, four routes tested, and of the six whose measurements are clean, none above the interpolated static front—which is the strictest of the two baselines we could have used (§II), since it sits above the discrete static points. (Four were measured while other jobs shared the host; we re-measured one, which moved 6.5 pp closer without crossing, and the other three are still out.) We expected the reason to be a cancellation—admission policy controls how many requests are in flight, the per-step cost rises with that number, and the product is what the wall clock sees—and we built a two-part decomposition of the wall clock that makes the cancellation measurable. That decomposition holds, and the cost it measures is real, but it is not the whole reason and on the configuration our controllers run it is not the binding one: there the elasticity stays below one across the operating range, so added concurrency was still cheap and the engine had room. Our controllers did not take it. What we can say at the end is narrower than what we set out to say: on this hardware and this workload, the admission rules we could build and tune did not beat a static cap, the capacity estimate they act on is $1.79\times$ the truth, and a structural idle component that no policy we built recovers at a profit remains. The regime in which adaptivity should have paid—where the same switch is worth $+51\%$ and $-42\%$—is one where the controller is not dominated by any static point we measured, and does not dominate one either. The measurement we offer is a diagnostic an engineer can take before committing to a sweep; whether it predicts the outcome on another workload is a question we tested once and could not confirm.

The part we would carry elsewhere is not the negative result but the mechanism behind it. Prefix reuse is bounded by the workload—75.5% of our prompt tokens—and whether that bound survives added concurrency is set by the memory tier: device-only, realization falls from 92.5% to 36.4% as the running count rises, and the shortfall reappears as recomputation; with a host tier the static caps hold at 90.2–92.4% across the range our controllers occupy, and the worst of the nineteen host-tier runs that completed this trace—the one that disables every throttle—still realizes 72.8%, where the nearest device-only point, the cap of four at $\overline{R}_{\text{busy}}$ 1.65, manages 60.7%—while the device cache misses *more*. Two configurations at the same running count and the same pool occupancy realize 79.0% and 91.3% of the bound, which is why we describe the tier rather than the occupancy as the variable. That claim is measured on one trace family, and the matched pair behind it ran on two cards a day apart; the controlled version—one card, back to back—is queued rather than done, and we do not extend the claim past what we measured—but it is the finding a system designer can act on, and it is the one we would test first on different hardware. Two boundaries come with it. On a second workload both of our measurements survive in direction and neither in level: that trace realizes less of its own bound at a lower occupancy, and its elasticity turns positive at a lower running count—reaching 1.0 there where the main trace reaches 1.15 at 1.53—so neither an occupancy threshold nor a concurrency threshold transfers. And the variable-load case, which is where a controller should have its best chance, is measured but not usable—its baselines were taken under host co-tenancy and we report it as open rather than as a result.

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TABLE I

THE RUNS WE COMPARE. *Conc.* IS THE TIME-WEIGHTED MEAN NUMBER OF *live* SESSIONS, WHICH IS NOT THE RUNNING-REQUEST COUNT $\overline{R}$ USED IN §III—A LIVE SESSION IS PARKED IN A TOOL WAIT MOST OF THE TIME, SO `CAP5` SHOWS 4.89 LIVE AGAINST 1.84 RUNNING. *THE NO-CONTROL ARM WRITES NO ADMISSION LOG, SO ITS *Conc.* IS THE MEAN NUMBER OF LIVE SESSIONS TAKEN FROM THE PER-STEP LOG'S SESSION SPANS; ON THE FIVE STATIC CAPS THAT ESTIMATOR AGREES WITH THE ADMISSION-LOG ONE TO TWO DECIMALS (AND THE TWO 25-SESSION CAPS TOO). $\Delta$ IS THE SIGNED DISTANCE FROM THE STATIC ENVELOPE AT THAT WALL CLOCK; ABOVE $+1.0$ PP COUNTS AS A WIN, BELOW $-1.0$ PP AS A LOSS, AND THE BAND BETWEEN AS A TIE (§II). TEN FURTHER RUNS ON THIS TRACE COMPLETED BUT ARE NOT LISTED, BECAUSE THEY BELONG TO SUPERSEDED SWEEPS OR EARLIER HARNESS GENERATIONS: THE PRE-HICACHE BASE RUN, THE BUDGET-FRACTION AND RATE-LIMIT ARMS OF §V, AN EARLIER CAP-3 RUN, AND THE TWO MATCHED-PAIR ARMS WHOSE NUMBERS §V CITES AS EXTRA REPEATS OF THE BASE CONFIGURATION. WHERE WE CITE ANY OF THEM WE NAME THEM. FOUR RUNS STOPPED SHORT OF 999 (`HZ0` 995, `HZ2` 997, `RG4` 998, AND THE NO-CONTROL BASELINE 997); THEY ARE LISTED WITH THEIR MEASURED VALUES. OUR VERDICT SCRIPT EXCLUDES RUNS BELOW 999 STEPS FROM ITS VERDICT TABLE, SO THESE FOUR ARE NOT JUDGED AGAINST THE FRONT THERE; THEIR $\Delta$ HERE IS COMPUTED FROM THE SAME FULL-PRECISION COORDINATES AS EVERY OTHER ROW (`RUNS_PRECISE.CSV` RECORDS THE STEP COUNT OF EACH RUN), OVER A WINDOW UP TO FOUR STEPS SHORTER. ‡ MARKS A RUN MEASURED WHILE THREE OR MORE OTHER ENGINE INSTANCES SHARED THE HOST FOR AT LEAST AN HOUR. THAT IS A RISK FLAG, NOT A MEASUREMENT OF DAMAGE, AND THE TWO COME APART IN OUR DATA. FOUR ROWS CARRIED THE FLAG AND THREE ARE DAMAGED: `HORIZON`$=2$, `HORIZON`$=0$ AND `HORIZON`$=1$ NO HW ARE CONTROLLER VARIANTS WHOSE PER-REQUEST DECODE INTERVAL IS 30.9–33.3 MS AGAINST 23.7–25.0 ACROSS THE SEVEN CLEAN RUNS OF THE SAME CONTROLLER FAMILY, SO THEIR WALL CLOCK, SLO AND $\Delta$ ARE NOT COMPARABLE WITH THE REST, AND THEY ARE BEING RE-MEASURED. THE FOURTH, THE RESIDENT BASE, HAS BEEN: ITS ROW ABOVE CARRIES THE CLEAN NUMBERS (24.8 MS $\tau_2$, 25.4 MINUTES EARLIER, $\Delta$ $-1.33$ AGAINST THE BASE CONFIGURATION'S $-1.24$). THE TWO REMAINING FLAGGED ROWS, `CAP1` AND THE PUBLISHED-PARAMETER AIMD ROW, SHOW NO AGGREGATE INFLATION AT ALL (20.6 AND 24.4 MS), SO WE KEEP THEIR NUMBERS AND SAY WHY. §IX-0E GIVES THE AUDIT, THE THRESHOLD AND THE EVIDENCE.

| Policy | Wall (min) | Conc. | SLO (%) | $\Delta$ (pp) |
|---|---|---|---|---|
| *Static caps* | | | | |
| no control* | 288.6 | 94.9 | 3.0 | −87.62 |
| cap1‡ | 527.4 | 1.00 | 94.8 | *anchor* |
| cap2 | 304.2 | 1.99 | 89.5 | −2.23 |
| cap3 (fixed) | 258.7 | 2.89 | 74.4 | −11.77 |
| cap4 | 249.9 | 3.94 | 55.7 | −29.17 |
| cap5 | 260.9 | 4.89 | 27.1 | −59.34 |
| *Static caps + HiCache* | | | | |
| cap2+HC | 295.0 | 1.99 | 91.6 | *anchor* |
| cap3+HC | 236.4 | 2.93 | 81.1 | −1.72 |
| cap3+HC (repeat) | 230.3 | 2.97 | 81.9 | *anchor* |
| cap4+HC | 213.0 | 3.93 | 63.4 | −7.55 |
| cap4+HC (repeat) | 209.1 | 3.87 | 68.5 | *anchor* |
| *Foyer controller variants* | | | | |
| single-flight off | 263.3 | 2.43 | 86.3 | −0.55 |
| horizon=1 | 255.8 | 2.58 | 84.8 | −0.93 |
| horizon=3 (base) | 291.3 | 2.06 | 89.8 | −1.24 |
| horizon=3 (2nd run) | 293.2 | 2.07 | 89.6 | −1.73 |
| horizon=2‡ | 306.1 | 2.22 | 86.9 | −4.81 |
| horizon=1 (repeat) | 260.8 | 2.69 | 80.9 | −5.58 |
| resident base | 265.9 | 2.41 | 85.9 | −1.33 |
| horizon=0‡ | 290.0 | 2.69 | 77.2 | −13.60 |
| horizon=1, no hw‡ | 310.9 | 2.77 | 72.3 | −19.54 |
| no gate, no resv. | 272.3 | 3.25 | 66.6 | −21.62 |
| running gate=4 | 282.3 | 3.03 | 64.3 | −25.39 |
| all throttles off | 415.0 | 2.58 | 46.1 | −47.10 |
| *Independent adaptive controller (§V-G)* | | | | |
| AIMD, paper params‡ | 364.3 | 1.91 | 81.4 | −11.16 |
| AIMD, retuned | 264.6 | 3.87 | 52.8 | −34.28 |

TABLE II

THE 25-SESSION TIER, ALL FOUR ARMS UNDER THE CURRENT HARNESS. THIS IS THE SAME-GENERATION COMPARISON EARLIER DRAFTS OF THIS PAPER WERE MISSING; `cap4` AND THE FOYER ARM WERE RUN FOR THIS REVISION. THE SEGMENT SD HERE IS OVER SIX SEGMENTS, NOT THE TEN TABLE I USES FOR 999-STEP RUNS, SO THE TWO TABLES' SD COLUMNS ARE NOT COMPARABLE.

| Policy | Wall (min) | Conc. | SLO (%) | Seg. SD (pp) |
|---|---|---|---|---|
| cap2 | 32.4 | 1.95 | 92.8 | 11.0 |
| cap3 | 24.2 | 2.77 | 91.2 | 5.1 |
| cap4 | 23.6 | 3.39 | 79.2 | 21.4 |
| Foyer | 27.6 | 2.47 | 92.8 | 7.1 |